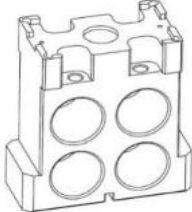
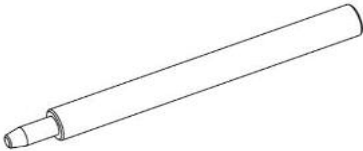
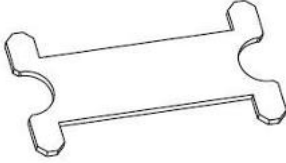
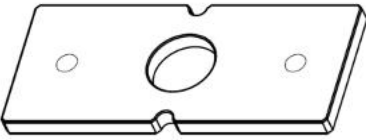


Reagent Manager Position and Gripper Assy Adjustment

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1. Adjustment tool

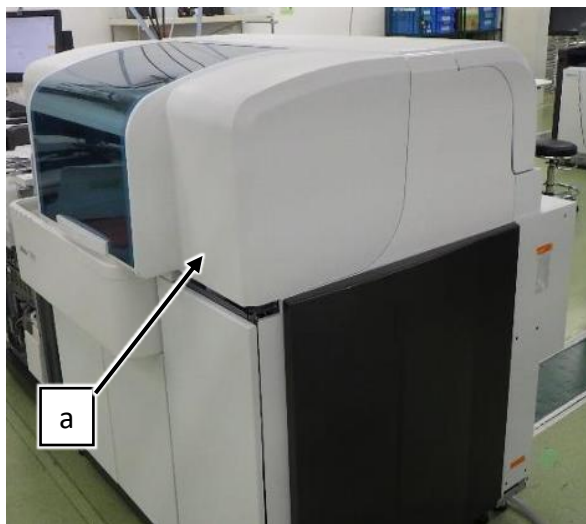
The following adjustment tools are used for this adjustment. Please refer to "S03-01 List of adjustment tools" for a list of jigs.

P/N	Name	Appearance
792-6455	GRIPPER ADJUSTING TOOL 1	
792-6457	GRIPPER ADJUSTING TOOL 2	
848-3760	PIERCE ADJUSTING TOOL	
848-3761	PIERCE Z ADJUSTING PLATE	
792-6453	BOTTLE PLATE C	

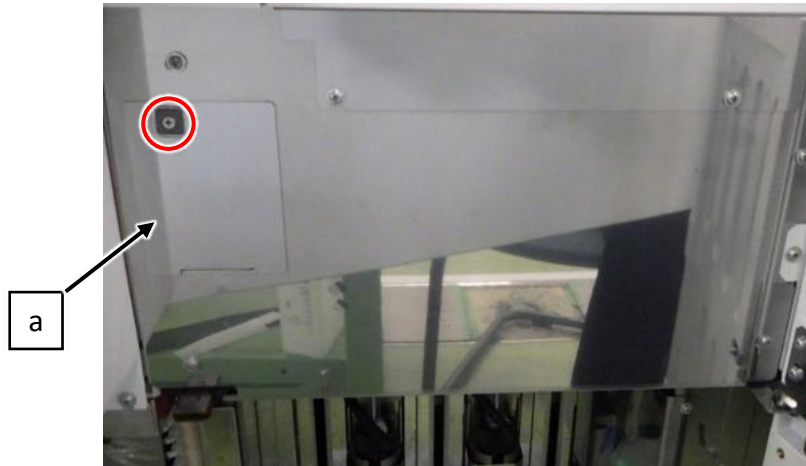
2. Preparation

2.1 Removing the covers

- (1) Open COVER TOP F ASSY (a).



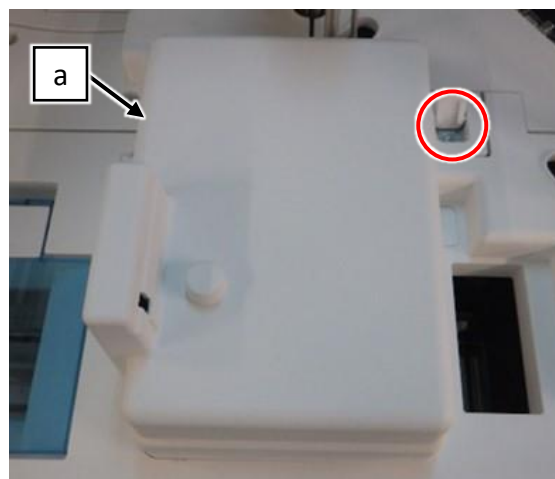
- (2) Loosen the screw and remove COVER MSW ASSY (a).



- (3) Turn on Maintenance switch.



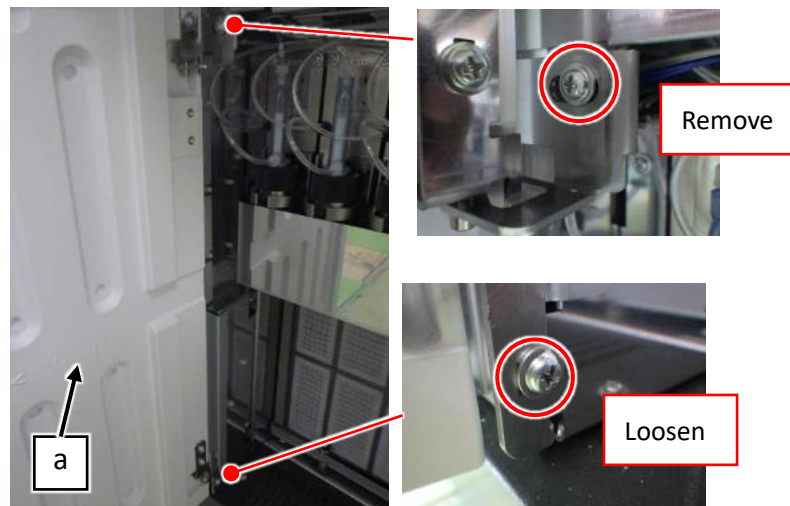
- (4) Remove the screw and COVER PIERCE UNIT (a).



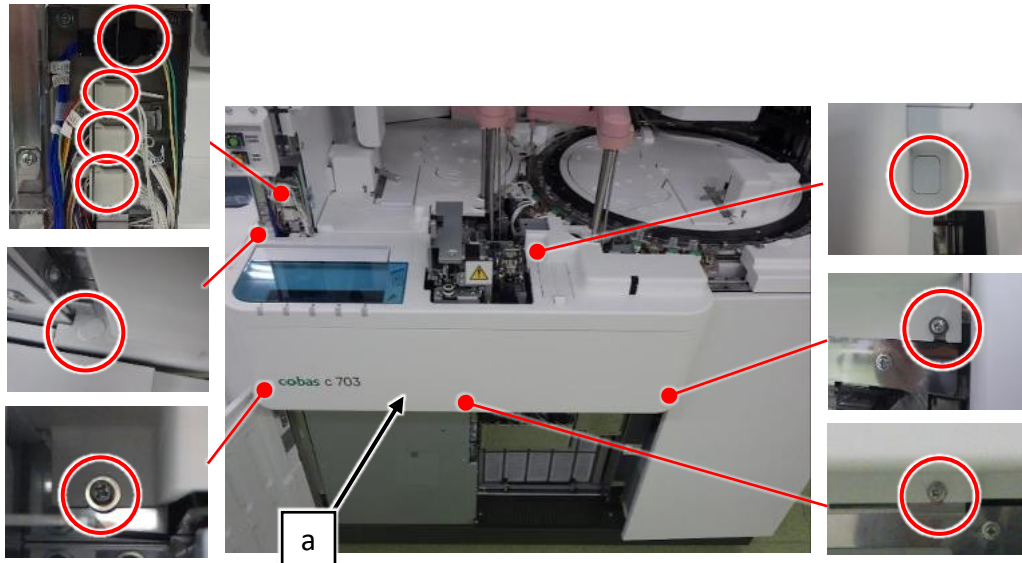
- (5) Loosen the 2 screws and remove SW PANEL ASSY (a).



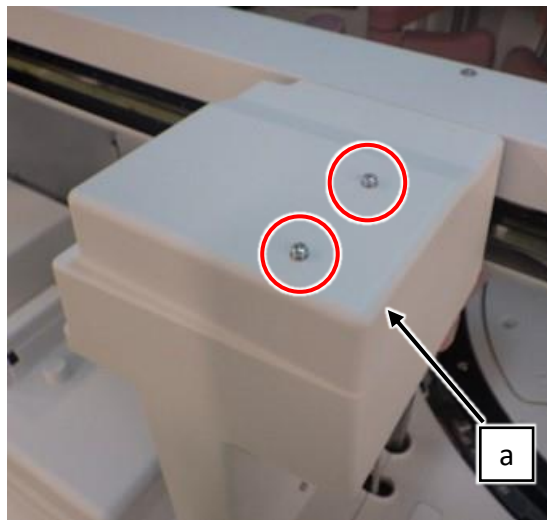
- (6) Loosen the screw and remove the screw then open FRONT DOOR (a).



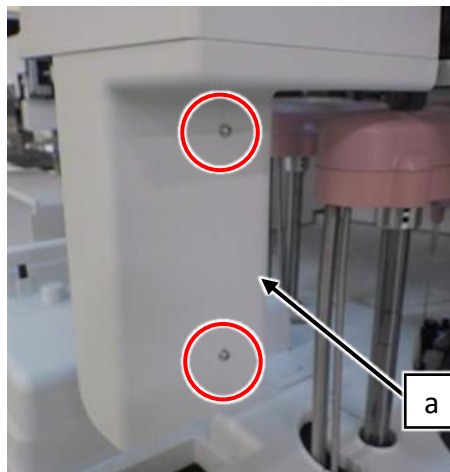
- (7) Remove the 5 screws and COVER TRAY TRANSPORT (a). When removing the cover, be careful NOT to forget to disconnect the 4 connectors for the motor and sensor at the front left.



- (8) Remove the 2 screws and COVER GRIPPER Z 1 (a).



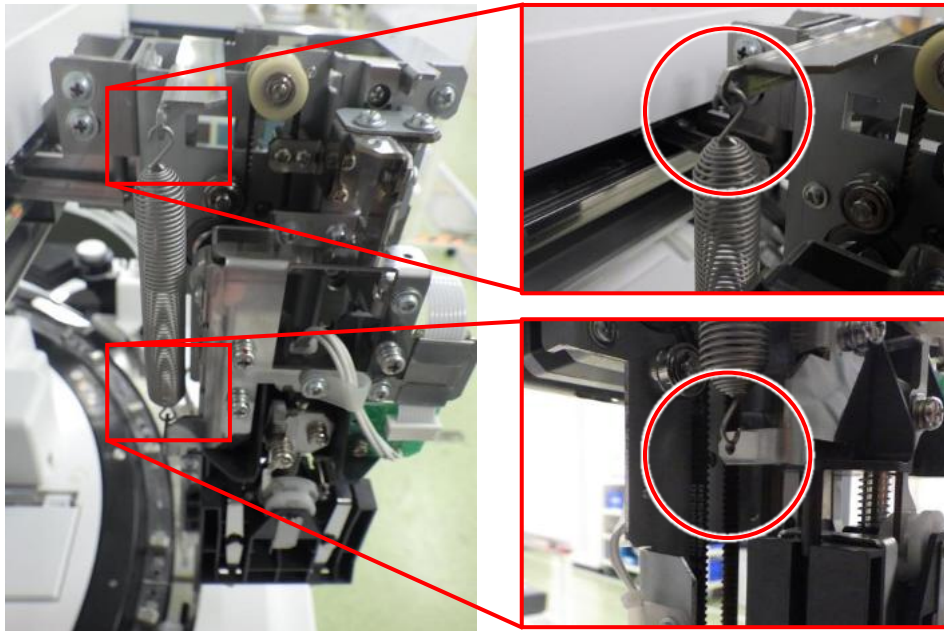
- (9) Remove the 2 screws and COVER GRIPPER Z 2 (a).



3. Adjustment of Gripper Direction

Before starting adjustment by SSW, check Gripper direction according to the section 3.1.

- (1) Remove the spring of Gripper.

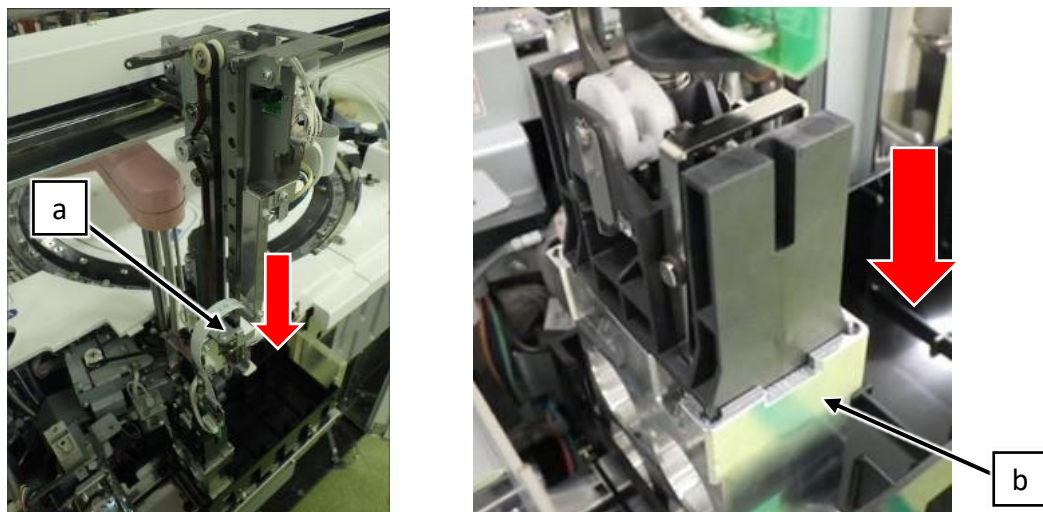


- (2) Drop the gripper under its own weight and make sure that Gripper head fits into the groove of the tool (b). If it does not drop, gently push the top of gripper (a) and wait for the gripper to fall.

Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.

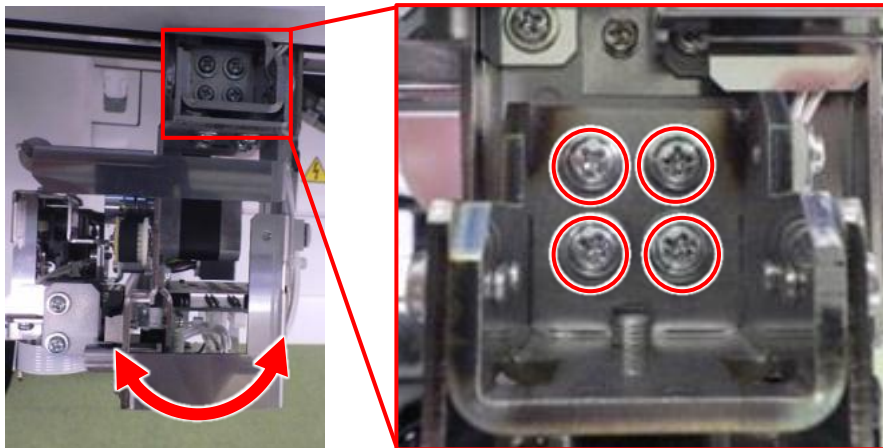
If it fits the groove smoothly, proceed to the step (8).

If not, proceed to the step (2).



(b): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (3) Loosen the 4 screws.



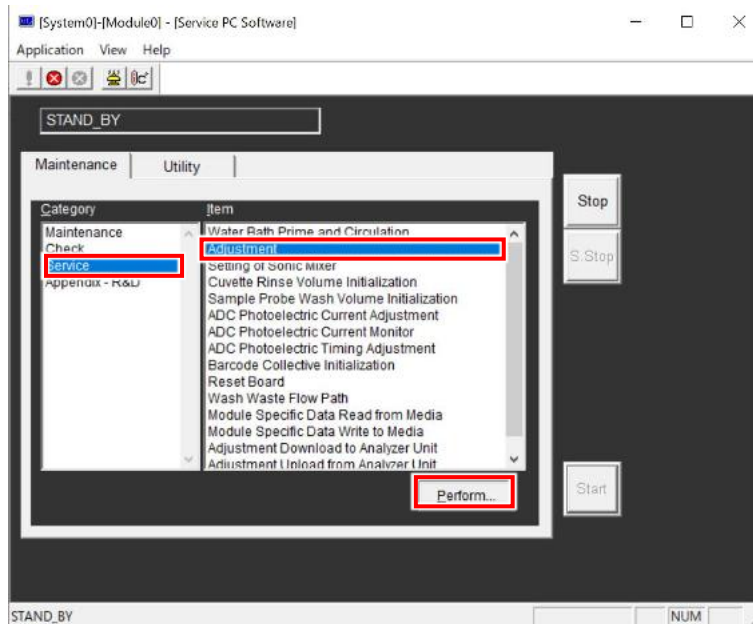
- (4) Adjust Gripper horizontal position so that Gripper head fits into the groove of the tool (b)
- (5) Tighten the 4 screws.
- (6) Lift Gripper head manually.
- (7) Retry from the step (2).
- (8) Remove the tool from the reagent loader.

4. Adjustment of Reagent Disk A

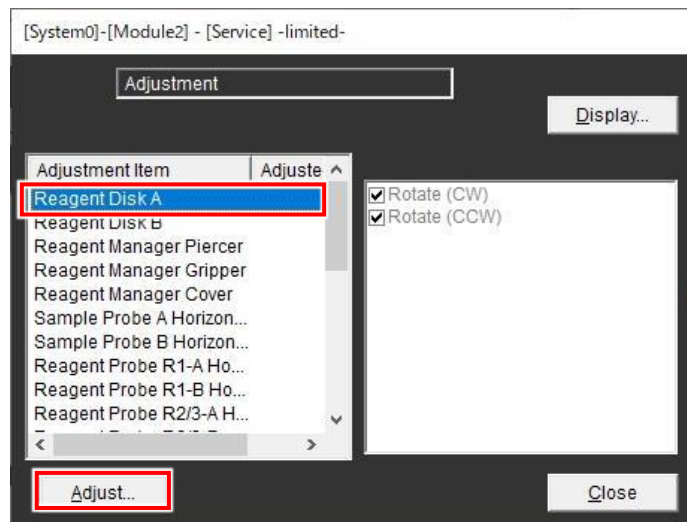
The adjustment procedure for Reagent Disk B is basically same as the procedure for Reagent Disk A.

4.1 Start SSW

- (1) Refer to “17 pro v2 SSW Adjustment Procedure”.
- (2) Select [Service] > [Adjustment] > [Perform...] in SSW.

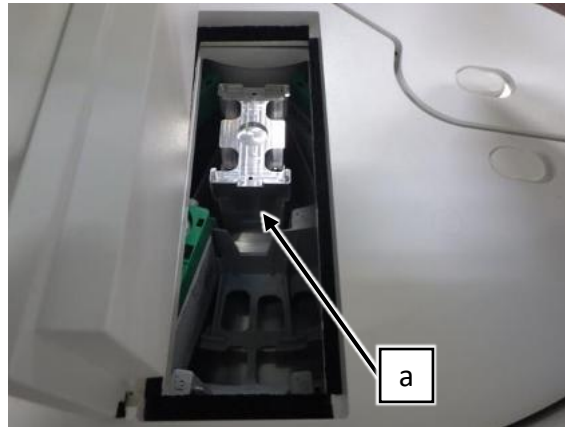


- (3) Select [Reagent Disk A] from the Adjustment item and select [Adjust...].



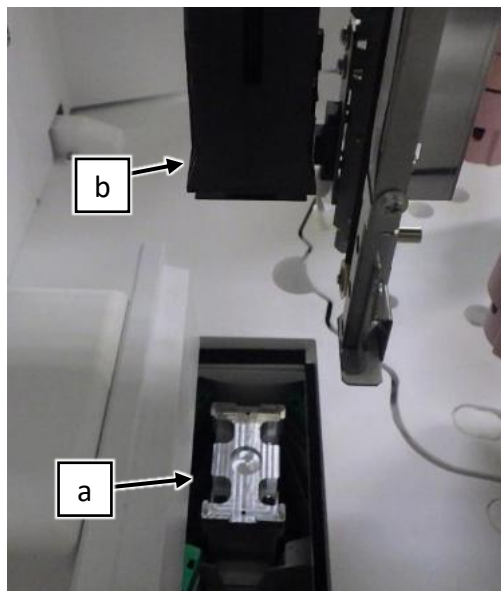
4.2 Rotate (CW) position

- (1) Set the tool (a) to Pos.1 of Reagent Disk A.



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (2) Move Gripper manually so that Gripper head (b) is above the tool (a).



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

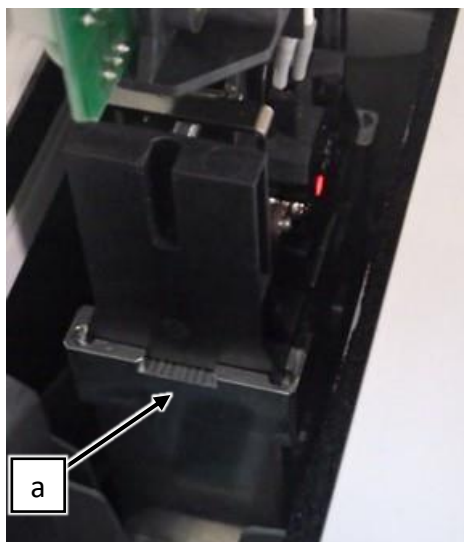
S03-08 Reagent Manager Position and Gripper Assy Adjustment

- (3) Drop the gripper under its own weight and make sure that Gripper head align in parallel to the groove of the tool (a). If it does not drop, gently push the top of gripper and wait for the gripper to fall.

Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.

If it fits the groove smoothly, proceed to the step (5).

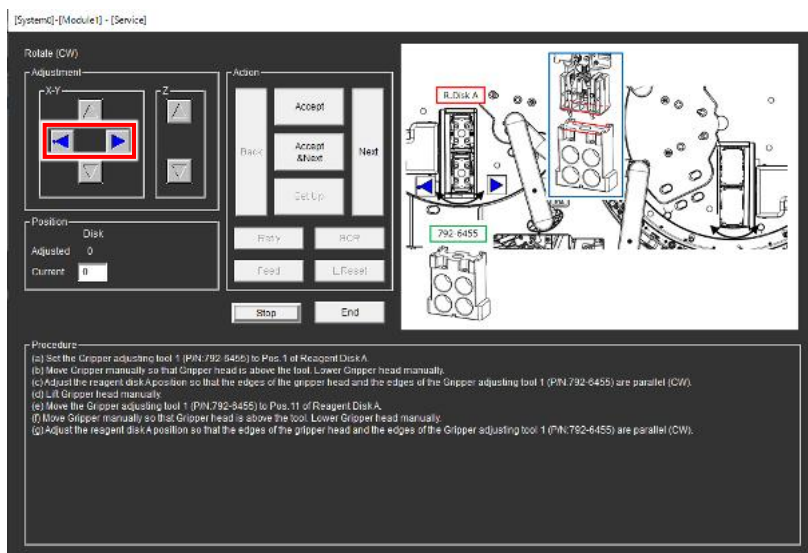
If not, proceed to the step (4).



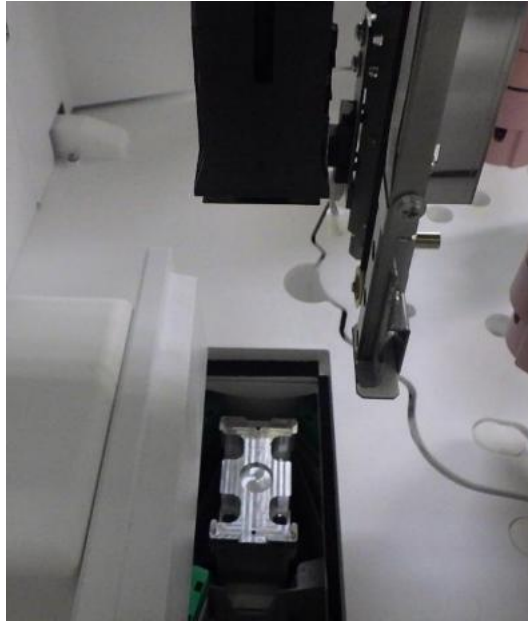
(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (4) Adjust the tool (a) position so that Gripper head fits smoothly into the tool (a) groove using the 2 arrow buttons.

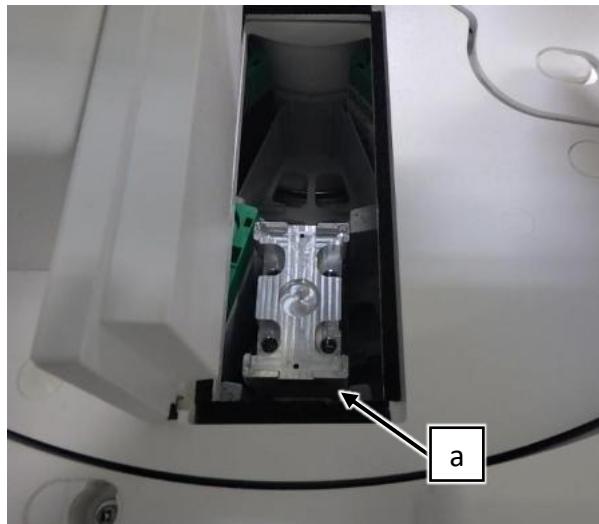
Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.



- (5) Lift Gripper head manually.

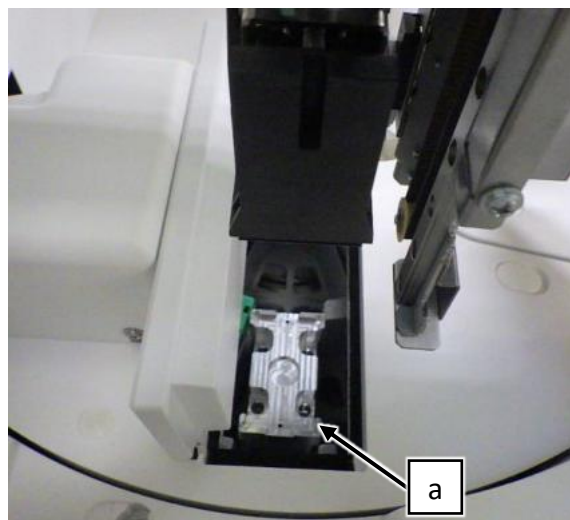


- (6) Move the tool (a) from Pos. 1 to Pos.11 of Reagent Disk A.



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (7) Move Gripper manually so that Gripper head is above the tool (a).



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (8) Drop the gripper under its own weight and make sure that Gripper head align in parallel to the groove of the tool

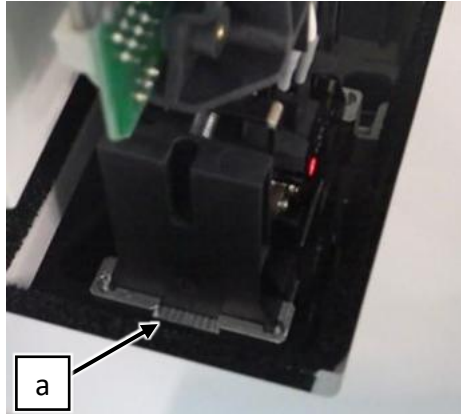
S03-08 Reagent Manager Position and Gripper Assy Adjustment

(a). If it does not drop, gently push the top of gripper and wait for the gripper to fall.

Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.

If it fits the groove smoothly, proceed to the step (10).

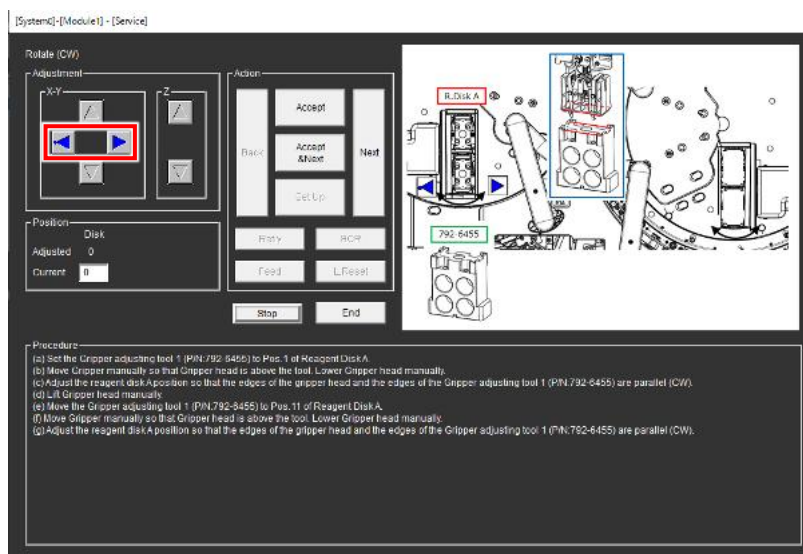
If not, proceed to the step (9).



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

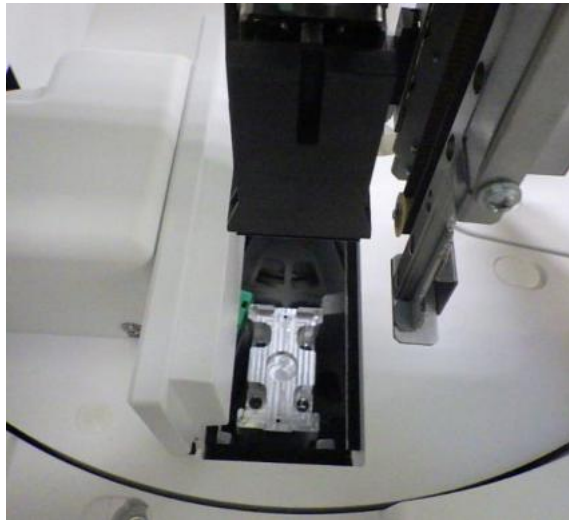
- (9) Adjust the direction of Reagent Disk A so that Gripper head fits smoothly into the tool (a) groove using the 2 arrow buttons.

Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.



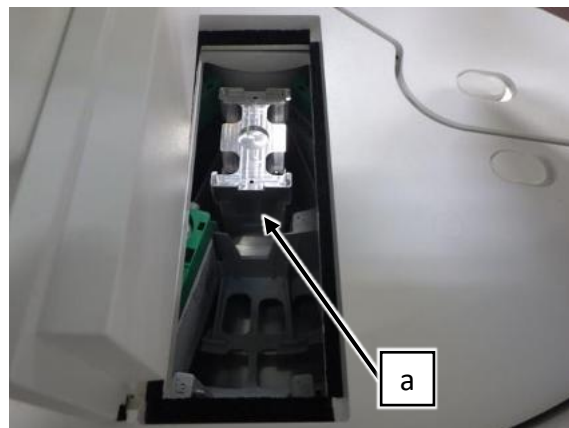
(10) Select [Accept & Next].

(11) Lift Gripper head manually.



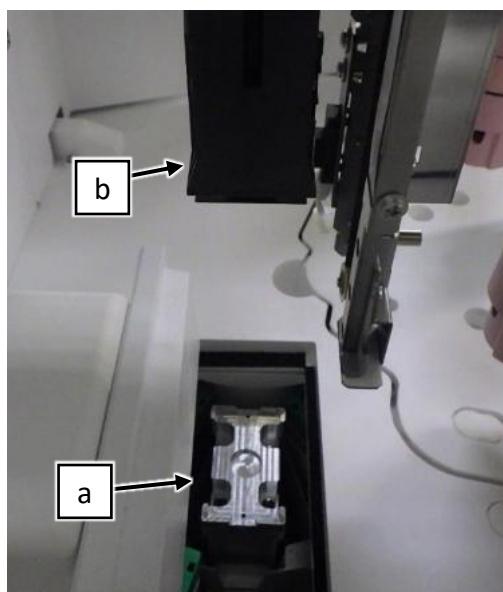
4.3 Rotate (CCW) position

(1) Move the tool (a) from Pos. 11 to Pos.1 of Reagent Disk A.



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

(2) Move Gripper manually so that Gripper head (b) is above the tool (a).



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

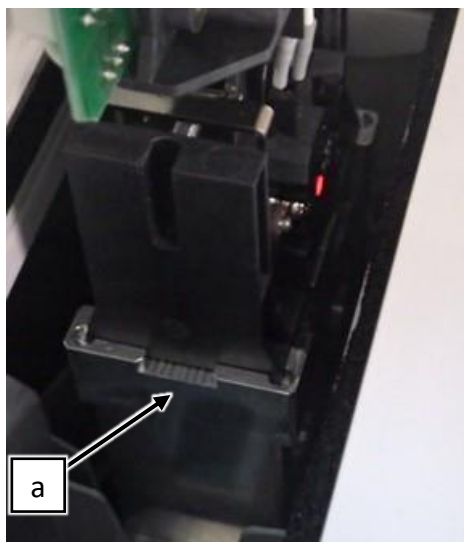
S03-08 Reagent Manager Position and Gripper Assy Adjustment

- (3) Drop the gripper under its own weight and make sure that Gripper head align in parallel to the groove of the tool
(a). If it does not drop, gently push the top of gripper and wait for the gripper to fall.

Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.

If it fits the groove smoothly, proceed to the step (5).

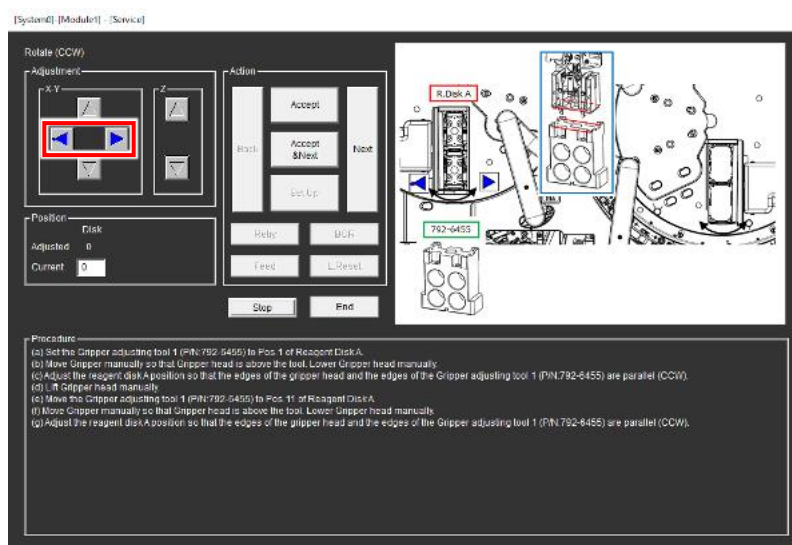
If not, proceed to the step (4).



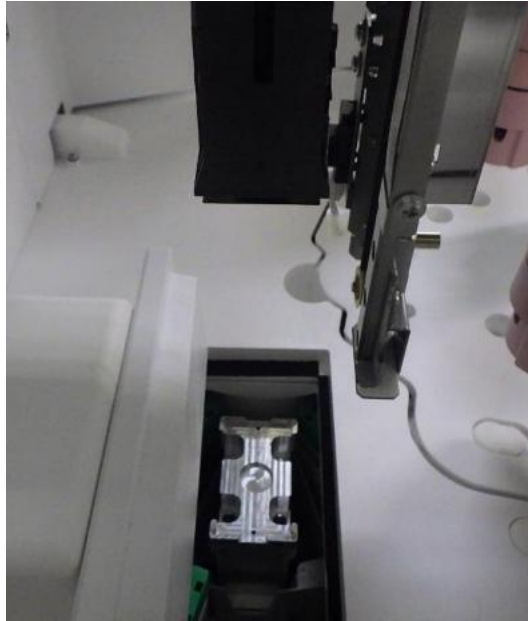
(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (4) Adjust the direction of Reagent Disk A so that Gripper head fits smoothly into the tool (a) groove using the 2 arrow buttons.

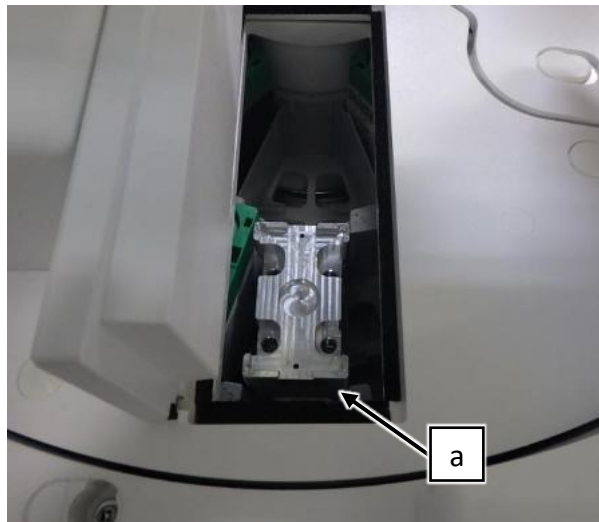
Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.



- (5) Lift Gripper head manually.

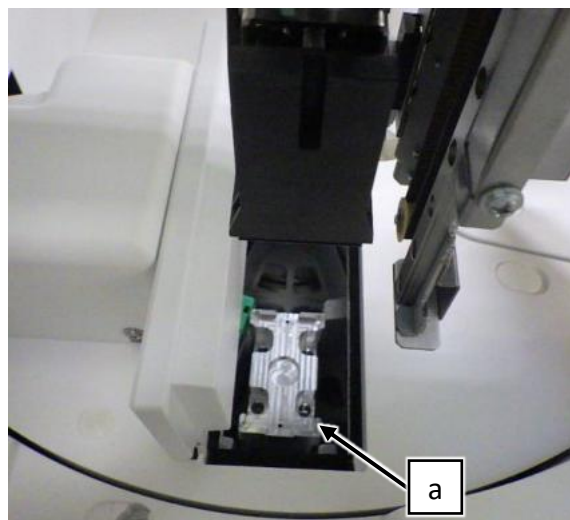


- (6) Move the tool (a) from Pos. 1 to Pos.11 of Reagent Disk A.



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (7) Move Gripper manually so that Gripper head is above the tool (a).



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

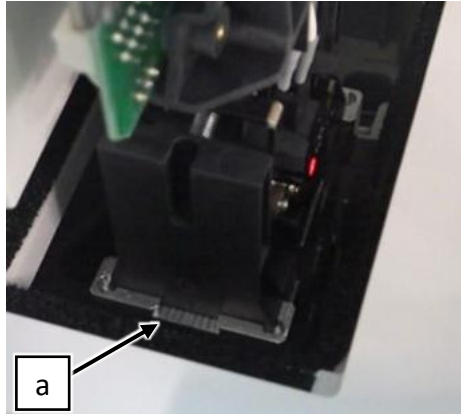
- (8) Drop the gripper under its own weight and make sure that Gripper head align in parallel to the groove of the tool

(a). If it does not drop, gently push the top of gripper and wait for the gripper to fall.

Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.

If it fits the groove smoothly, proceed to the step (10).

If not, proceed to the step (9).



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

(9) Adjust the direction of Reagent Disk A so that Gripper head fits smoothly into the tool (a) groove using the 2 arrow buttons.

Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.

[System01] [Module01] [Service]

Rotate (CCW)

Adjustment

A, Y

Left

Right

Position

Disk

Adjusted 0

Current 0

Action

Accept

Back

Accept & Next

Next

Set Up

Exit

MCU

End

L-Reset

Stop

End

Procedure

(a) Set the Gripper adjusting tool 1 (P/N:792-6455) to Pos. 1 of Reagent Disk A.

(b) Move Gripper manually so that Gripper head is above the tool. Lower Gripper head manually.

(c) Adjust the reagent disk A position so that the edges of the gripper head and the edges of the Gripper adjusting tool 1 (P/N:792-6455) are parallel (CCW).

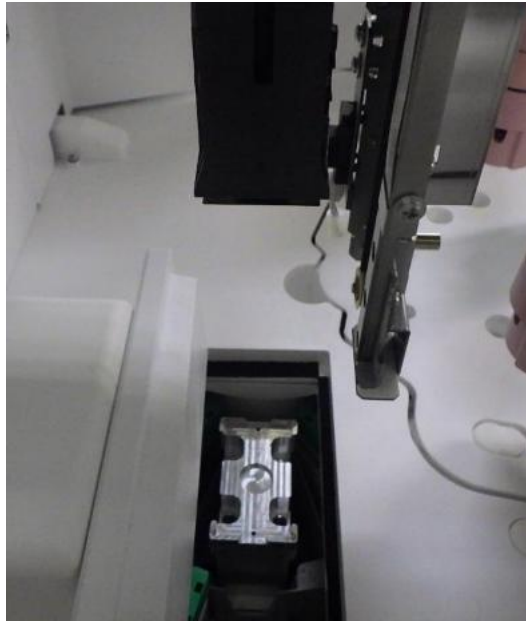
(d) L-Reset Gripper head manually.

(e) Move the Gripper adjusting tool 1 (P/N:792-6455) to Pos. 11 of Reagent Disk A.

(f) Move Gripper manually so that Gripper head is above the tool. Lower Gripper head manually.

(g) Adjust the reagent disk A position so that the edges of the gripper head and the edges of the Gripper adjusting tool 1 (P/N:792-6455) are parallel (CCW).

- (10) Select [Accept].
- (11) Lift Gripper head manually.



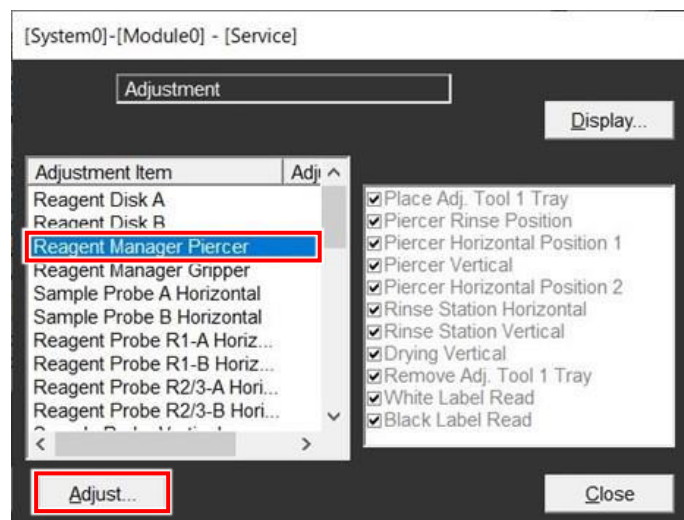
- (12) Remove the tool from Reagent disk A.
- (13) Select [End].

5. Adjustment of Reagent Disk B

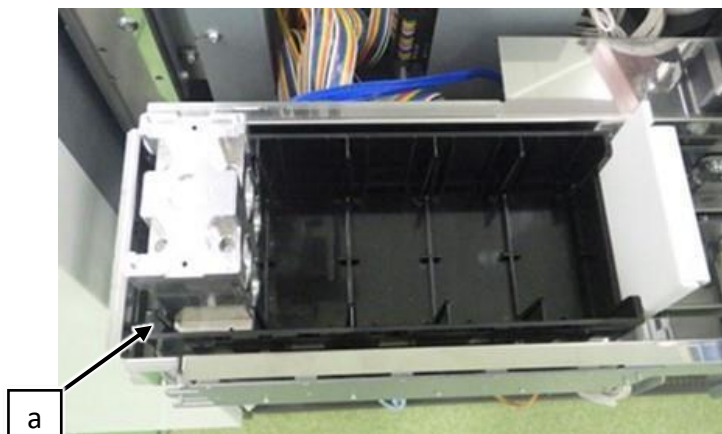
The adjustment procedure for Reagent Disk B is basically same as the procedure for Reagent Disk A.
Perform Reagent Disk B adjustment refer to the chapter 4.

6. Adjustment of Reagent Manager Piercer

- (1) Select [Reagent Manager Piercer] from Adjustment item and select [Adjust...].

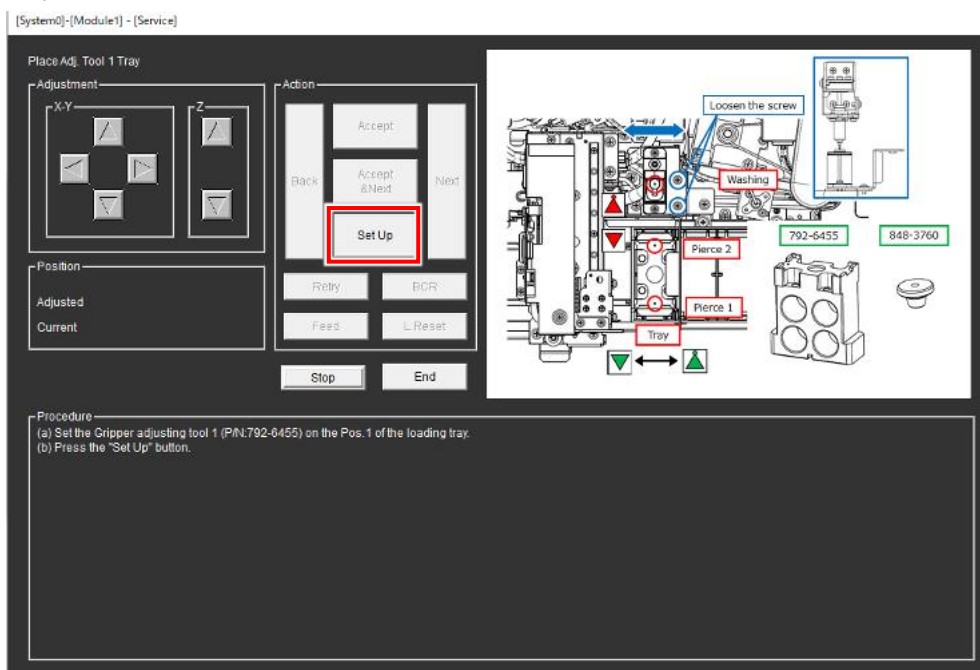


- (2) Set the tool (a) on Pos.1 of the reagent loader.



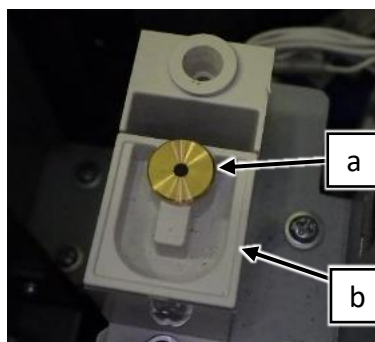
(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (3) Select [Set Up].



6.1 Piercer Rinse Position

- (1) Set the tool (a) in the front hole of PIERCE CLEANING BATH 3 (b).

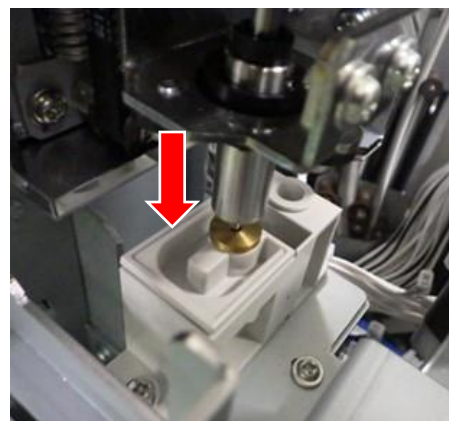
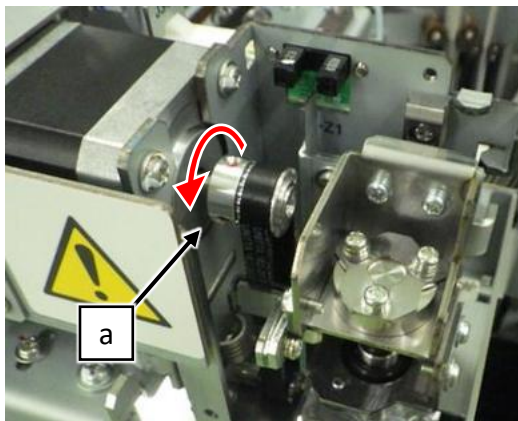


(a): PIERCE ADJUSTING TOOL (P/N:848-3760)

- (2) Move Piercer manually so that Piercer tip is above the tool.

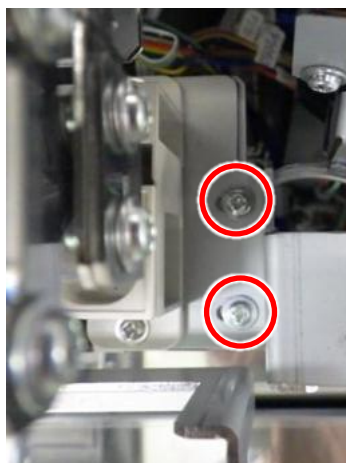


- (3) Rotate the pulley (a) to lower Piercer tip and make sure that Piercer tip is inserted into the hole of the tool.
Note that if Piercer tip itself is pulled when lowering, Piercer tip becomes an abnormal adjustment condition.
If it is inserted into the hole, proceed to the step (8).
If not, proceed to the step (4).



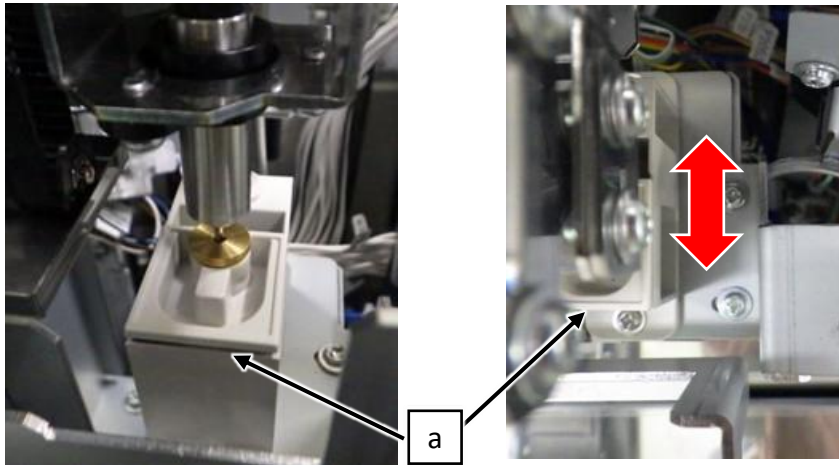
(a): PIERCE ADJUSTING TOOL (P/N:848-3760)

- (4) Loosen the 2 screws.



- (5) Rotate the pulley to lower Piercer tip and adjust PIERCE CLEANING BATH 3 (a) position so that Piercer tip is inserted into hole of the tool.

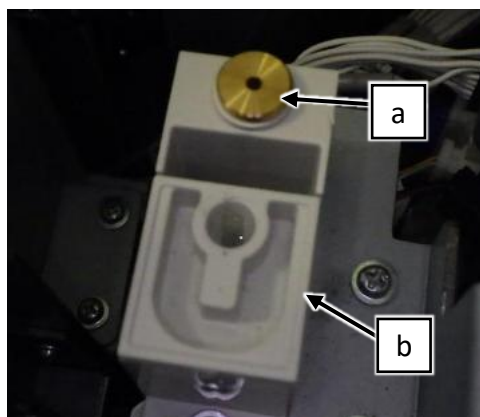
Note that if Piercer tip itself is pulled when lowering, Piercer tip becomes an abnormal adjustment condition.



- (6) Tighten the 2 screws.
(7) Lift Piercer tip.



- (8) Move the tool (a) to the rear hole of PIERCE CLEANING BATH 3 (b).



(a): PIERCE ADJUSTING TOOL (P/N:848-3760)

- (9) Move Piercer manually so that Piercer tip is above the tool.

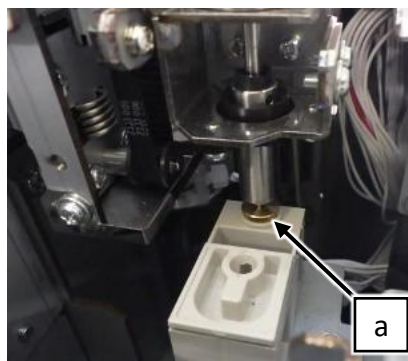


- (10) Rotate the pulley to lower Piercer tip and make sure that Piercer tip is inserted into hole of the tool (a).

Note that if Piercer tip itself is pulled when lowering, Piercer tip becomes an abnormal adjustment condition.

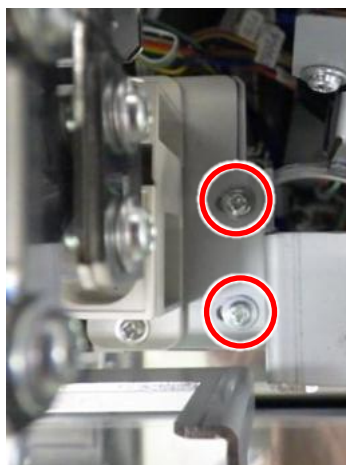
If it is inserted into the hole, proceed to the step (17).

If not, proceed to the step (11).



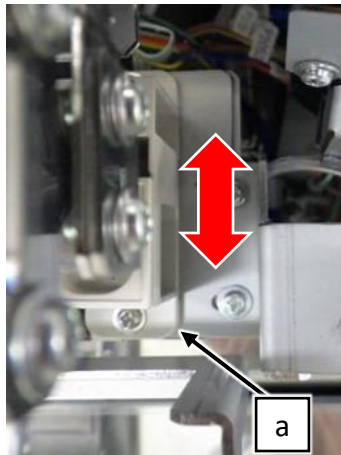
(a): PIERCE ADJUSTING TOOL (P/N:848-3760)

- (11) Loosen the 2 screws.



- (12) Rotate the pulley to lower Piercer tip and adjust PIERCE CLEANING BATH 3 (a) position so that Piercer tip is inserted into hole of the tool.

Note that if Piercer tip itself is pulled when lowering, Piercer tip becomes an abnormal adjustment condition.

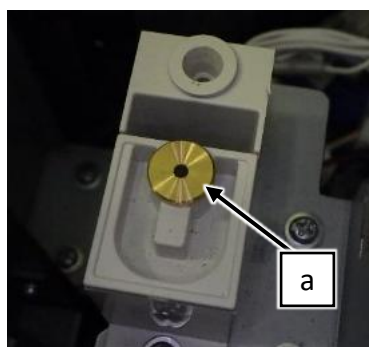


- (13) Tighten the 2 screws.

- (14) Lift Piercer tip.



- (15) Move the tool from the back hole to the front hole of PIERCE CLEANING BATH 3.



(a): PIERCE ADJUSTING TOOL (P/N:848-3760)

- (16) Retry from the step (2).

- (17) Remove the tool from PIERCE CLEANING BATH 3.

- (18) Select [Next].

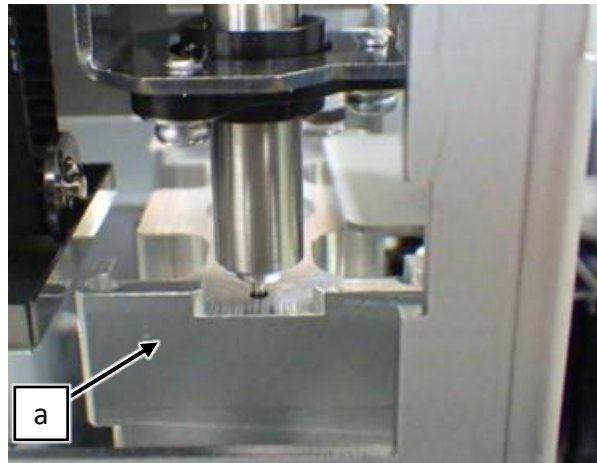
6.2 Piercer Horizontal Position 1

- (1) Rotate the pulley to lower Piercer tip and make sure that Piercer tip is inserted into hole of the tool (a).

Note that if Piercer tip itself is pulled when lowering, Piercer tip becomes an abnormal adjustment condition.

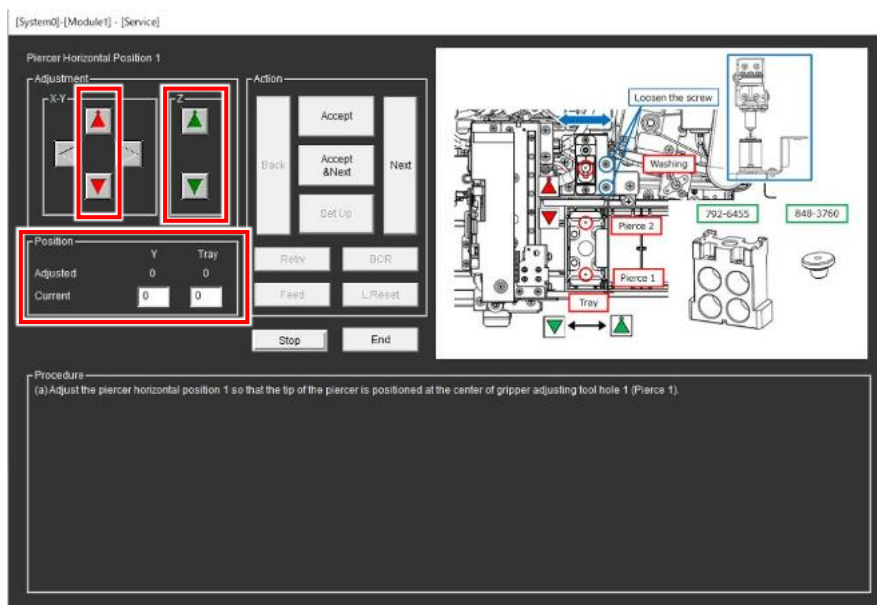
If it is inserted into hole, select [Next] and proceed to the section 6.3.

If not, proceed to the step (2).



(a): GRIPPER ADJUSTING TOOL 1 (P/N: 792-6455)

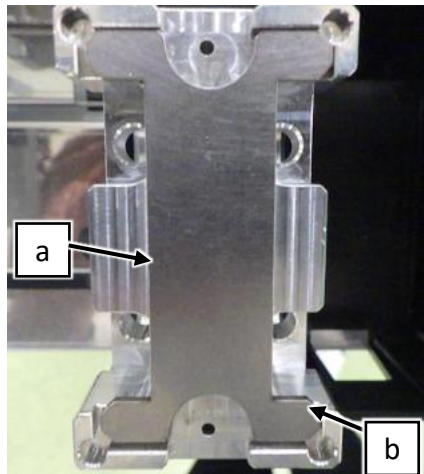
- (2) Adjust Piercer and the reagent loader position so that Piercer tip is inserted into the hole using the 4 arrow buttons.
(Range: Y: ± 30 pulse, Tray (the reagent loader): ± 20 pulse)



- (3) Select [Accept & Next].

6.3 Piercer Vertical

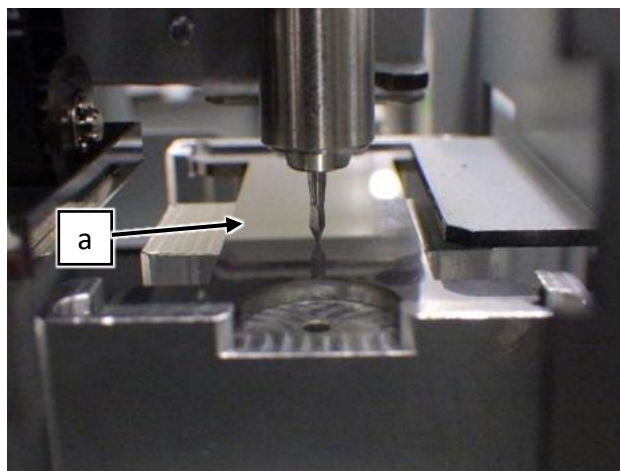
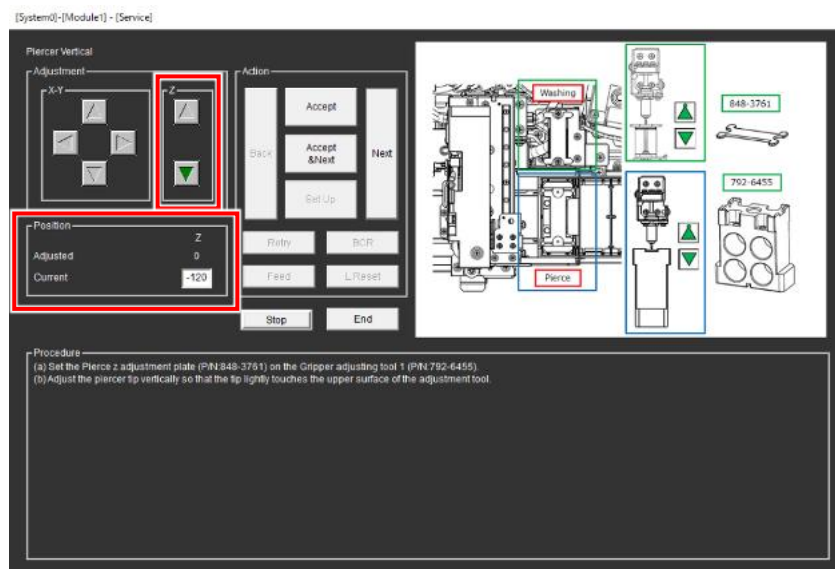
- (1) Set the plate (a) on the tool (b).



(a): PIERCE Z ADJUSTING PLATE (P/N: 848-3761)

(b): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (2) Adjust Piercer tip height so that Piercer tip touches the upper surface of the plate lightly using the 2 arrow buttons.
(Range: Z: ± 120 pulse)

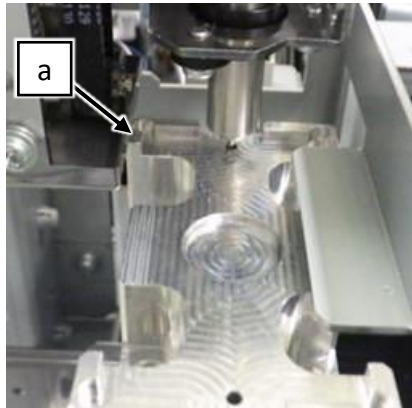


(a): PIERCE Z ADJUSTING PLATE (P/N: 848-3761)

- (3) Select [Accept & Next].

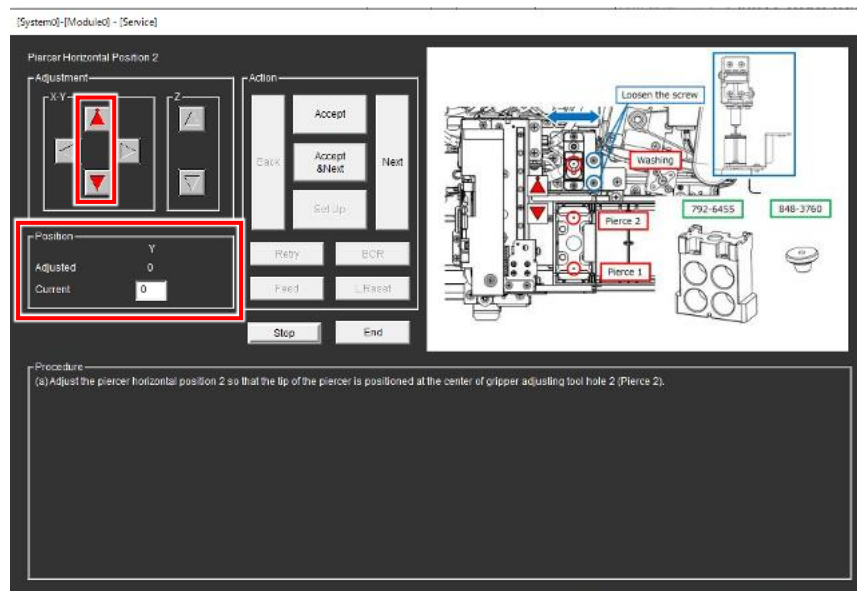
6.4 Piercer Horizontal Position 2

- (1) Rotate the pulley to lower Piercer tip and make sure that Piercer tip is inserted into the hole of the tool (a).
Note that if Piercer tip itself is pulled when lowering, Piercer tip becomes an abnormal adjustment condition.
If it is inserted into the hole, select [Next] and proceed to the section 6.5.
If not, proceed to the step (2).



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

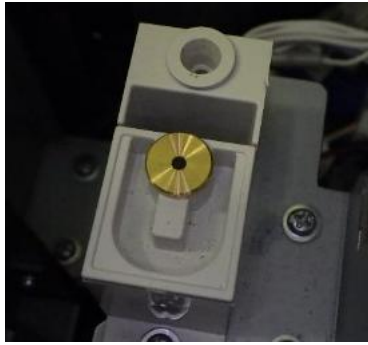
- (2) Adjust Piercer and the reagent loader position so that Piercer tip is inserted into the hole using the 2 arrow buttons.
Note that if Piercer tip itself is pulled when lowering, Piercer tip becomes an abnormal adjustment condition.
(Range: Y: ± 30 pulse)



- (3) Select [Accept & Next].

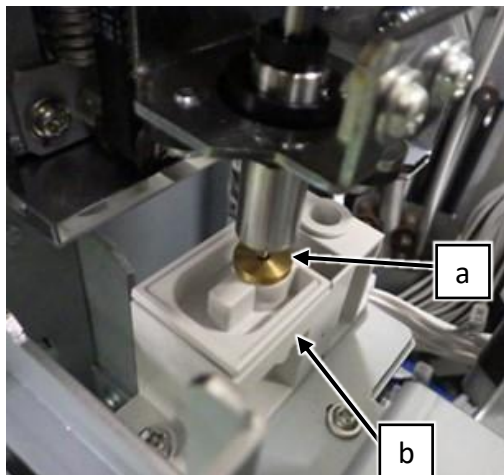
6.5 Rinse Station Horizontal

- (1) Set the tool (a) in the front hole of PIERCE CLEANING BATH 3 (b).



(a): PIERCE ADJUSTING TOOL (P/N:848-3760)

- (2) Rotate the pulley to lower Piercer tip and make sure that Piercer tip is inserted into hole of the tool.
Note that if Piercer tip itself is pulled when lowering, Piercer tip becomes an abnormal adjustment condition.
If it is inserted into hole, select [Next] and proceed to the section 6.6.
If not, proceed to the step (3)

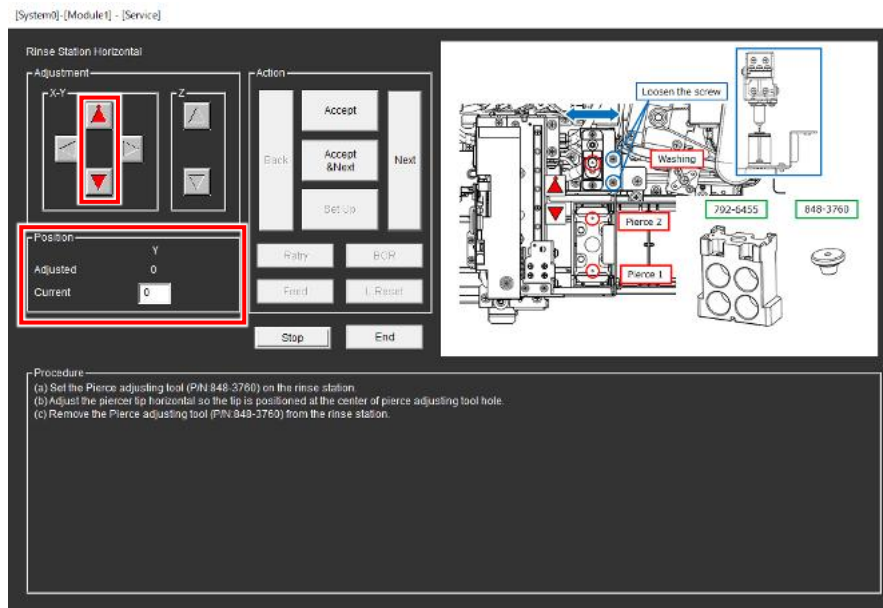


(a): PIERCE ADJUSTING TOOL (P/N:848-3760)

- (3) Adjust Piercer position so that Piercer tip is inserted into the hole using the 2 arrow buttons.

Note that if Piercer tip itself is pulled when lowering, Piercer tip becomes an abnormal adjustment condition.

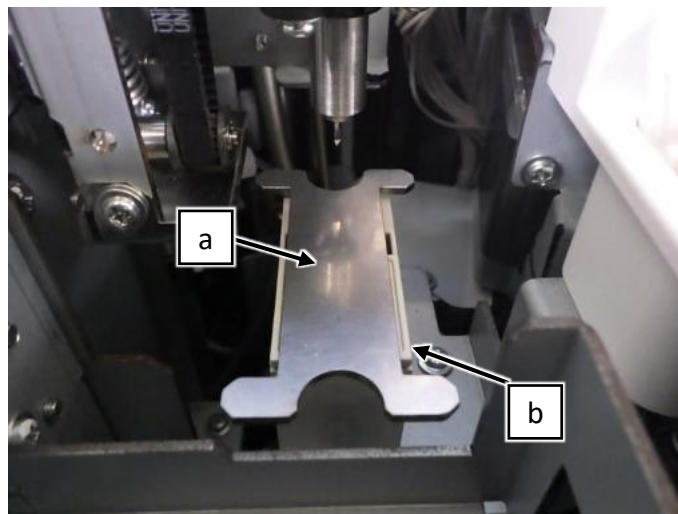
(Range: Y: ± 30 pulse)



- (4) Select [Accept & Next].

6.6 Rinse Station Vertical

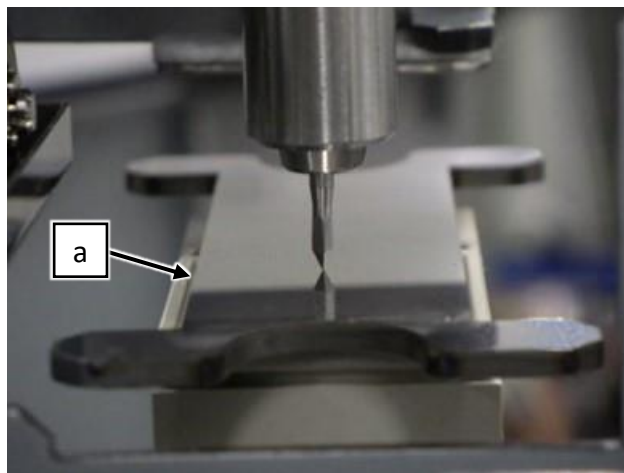
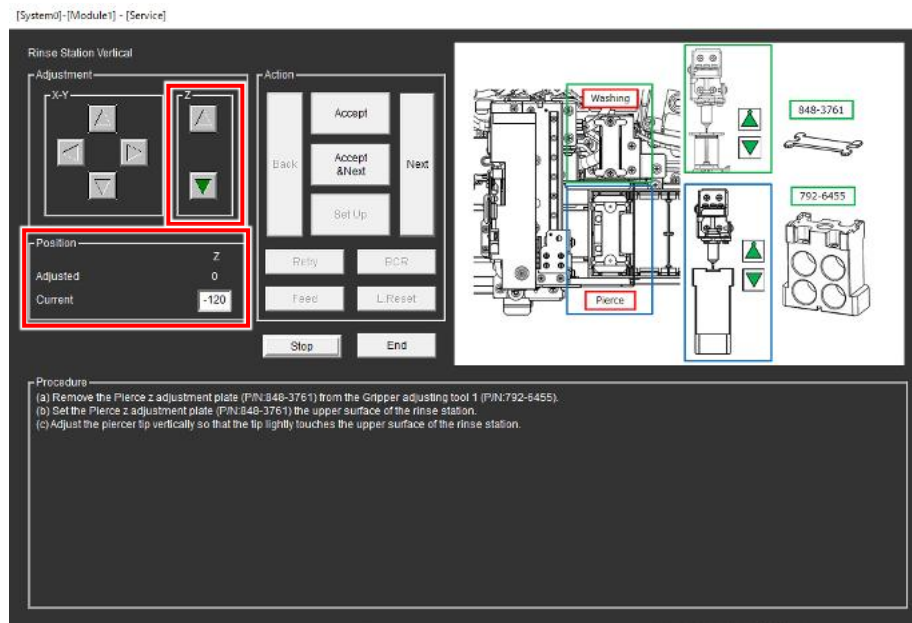
- (1) Set the plate (a) on PIERCE CLEANING BATH 3 (b).



(a): PIERCE Z ADJUSTING PLATE (P/N:848-3761)

(b): PIERCE CLEANING BATH 3

- (2) Adjust Piercer tip height so that Piercer tip touches the upper surface of the plate (a) lightly using the 2 arrow buttons.
(Range: Z: ± 120 pulse)

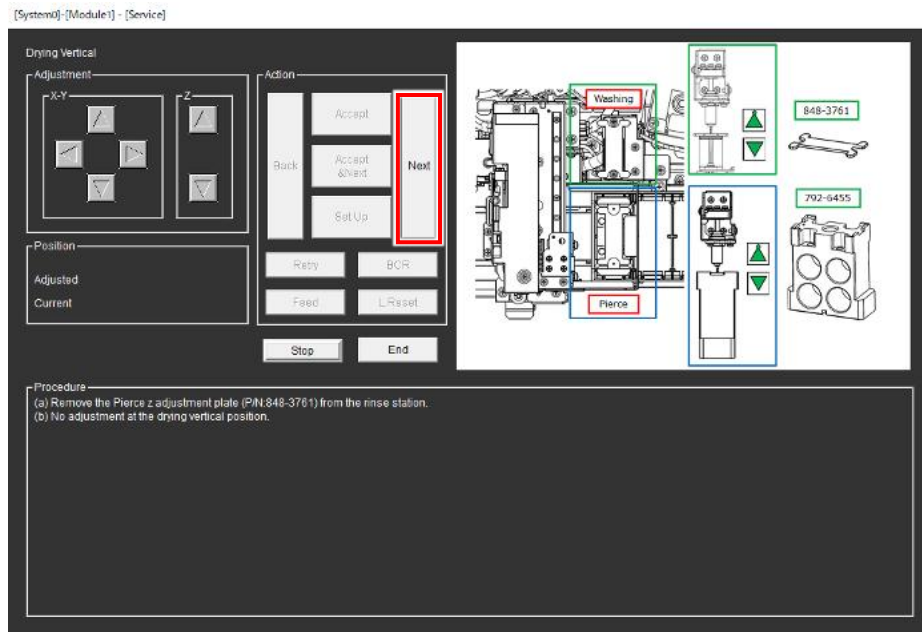


(a): PIERCE Z ADJUSTING PLATE (P/N:848-3761)

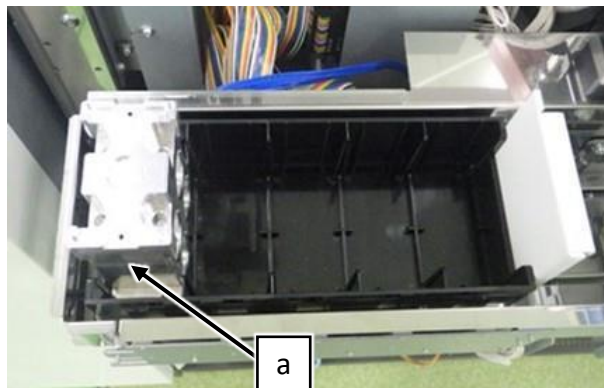
- (3) Select [Accept & Next].
(4) Remove the plate.

6.7 Drying Vertical

- (1) Select [Next].



- (2) Remove the tool (a) from the reagent loader.



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

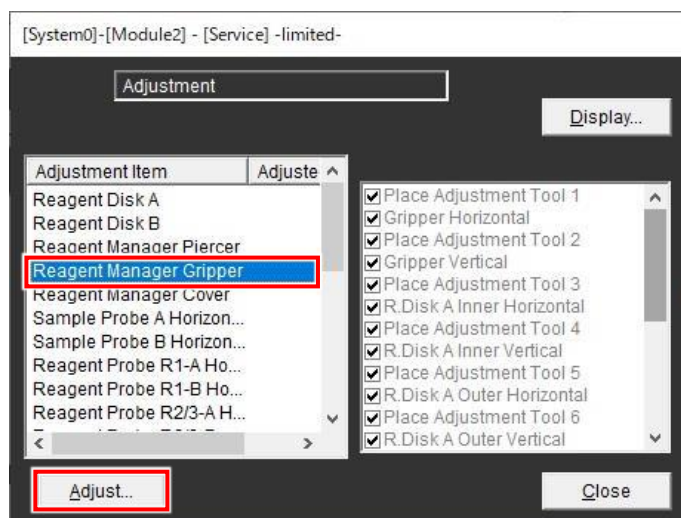
- (3) Select [End].

7. Adjustment of Reagent Manager Gripper

Make sure that the adjustments of the chapter 4 and 5 have been completed.

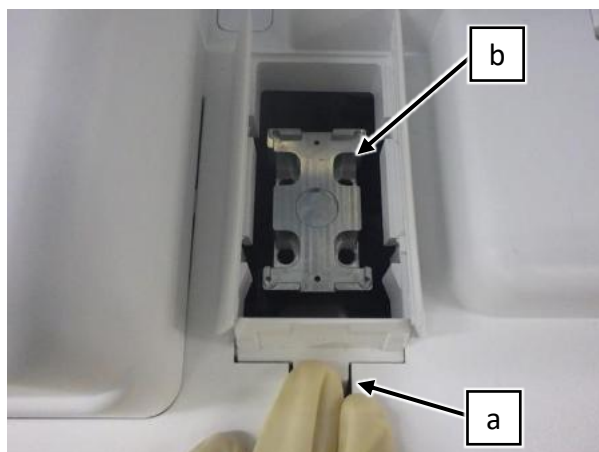
If they haven't been completed yet, perform the adjustments.

- (1) Attach COVER TRAY TRANSPORT. (Refer to the step 2.1 (7).)
- (2) Select [Reagent Manager Gripper] from Adjustment item and select [Adjust...].



7.1 Gripper Horizontal

- (1) Push the switch (a) and set the tool (b) to Pos.1 of the reagent loader.



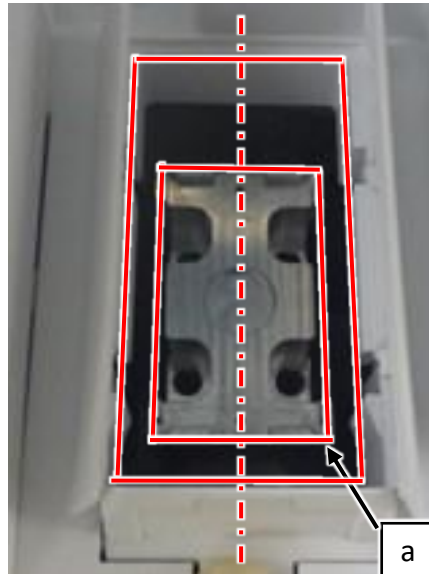
(b): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (2) Select [Set Up].

- (3) Make sure the tool (a) is at the center of the cover hole.

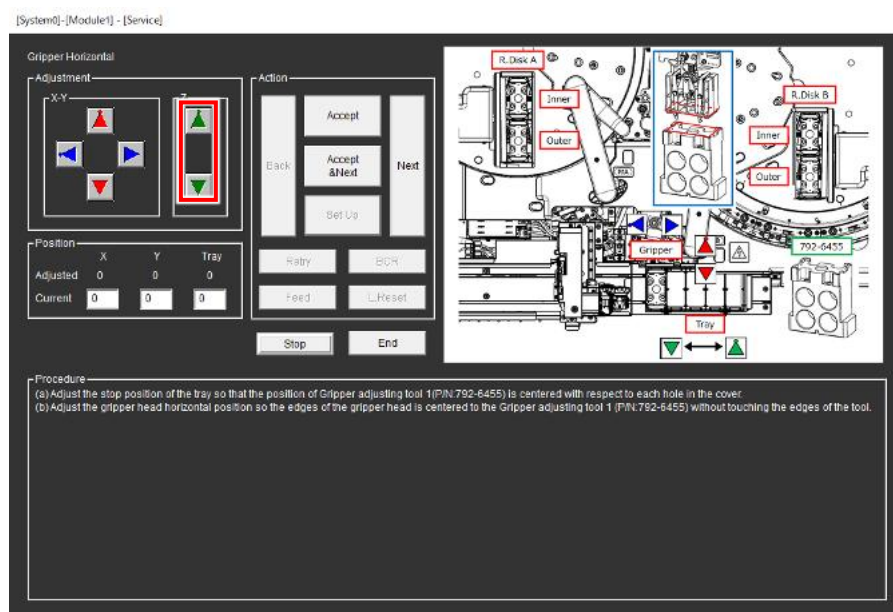
If it is at the center, proceed to step (6).

If not, proceed to step (4).



(a): GRIPPER ADJUSTING TOOL 1 (P/N: 792-6455)

- (4) Adjust the reagent loader position so that the tool is at the center of the cover hole using the 2 arrows button.



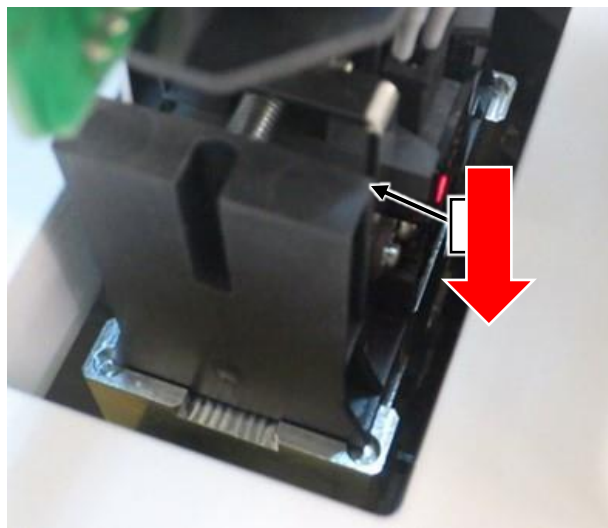
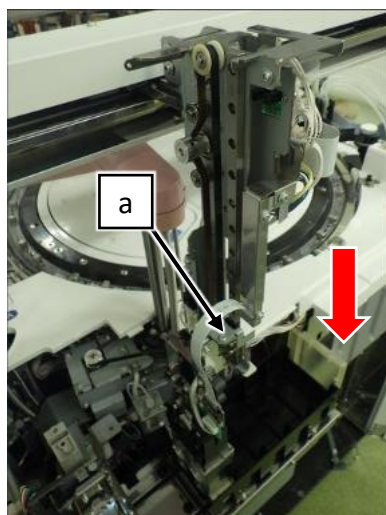
- (5) Select [Accept].

- (6) Drop the gripper under its own weight and make sure that Gripper head fits into the groove of the tool (b). If it does not drop, gently push the top of gripper (a) and wait for the gripper to fall.

Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.

If it fits smoothly, select [Next] and proceed to the section 7.2.

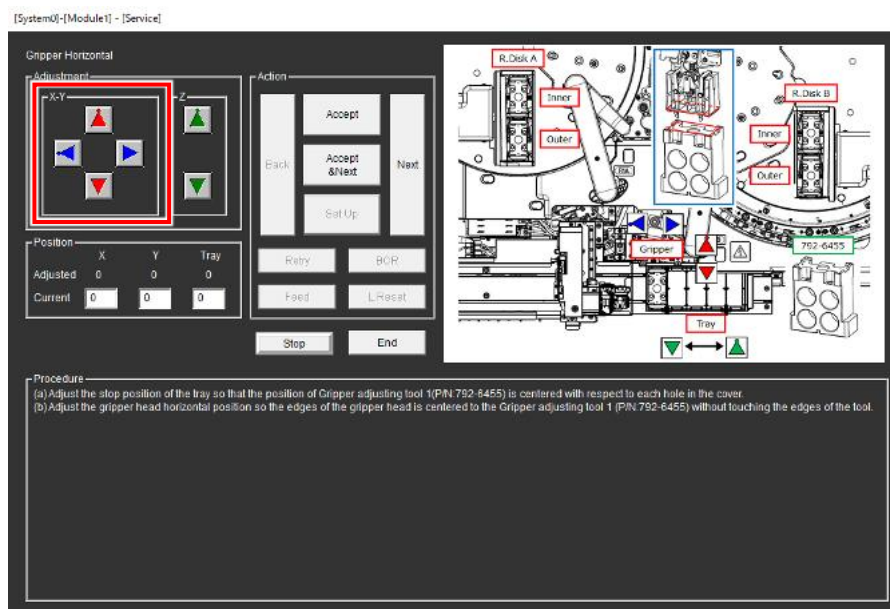
If not, proceed to the step (7).



(a): GRIPPER ADJUSTING TOOL 1 (P/N: 792-6455)

- (7) Adjust Gripper and the reagent loader position so that Gripper head fits smoothly into the tool groove using 4 arrow buttons.

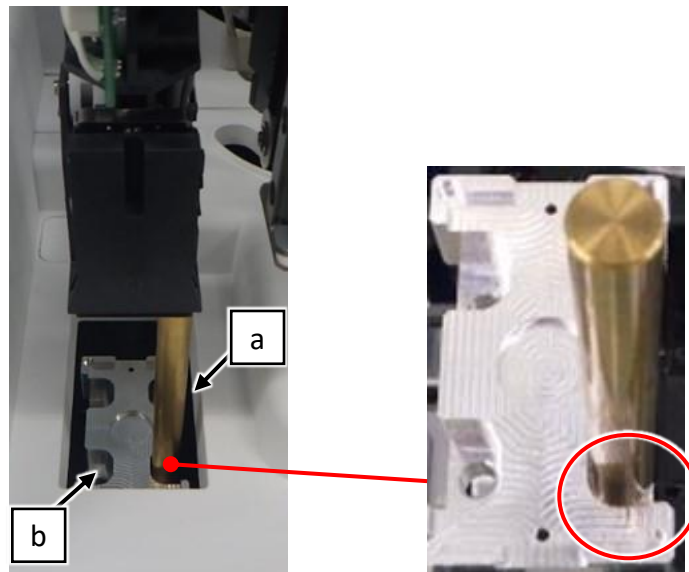
Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.



- (8) Select [Accept & Next].

7.2 Gripper Vertical

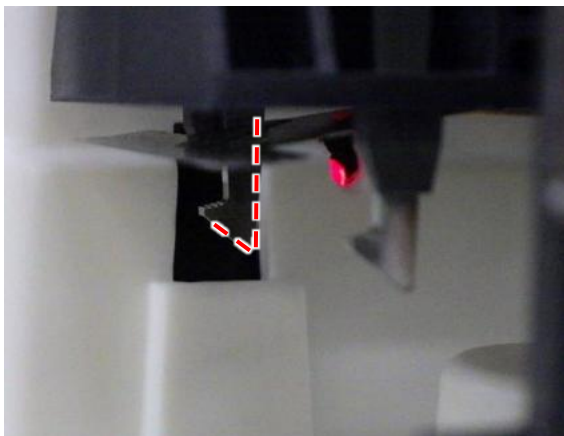
- (1) Set the pole (a) on the tool (b) hole in front of the right.



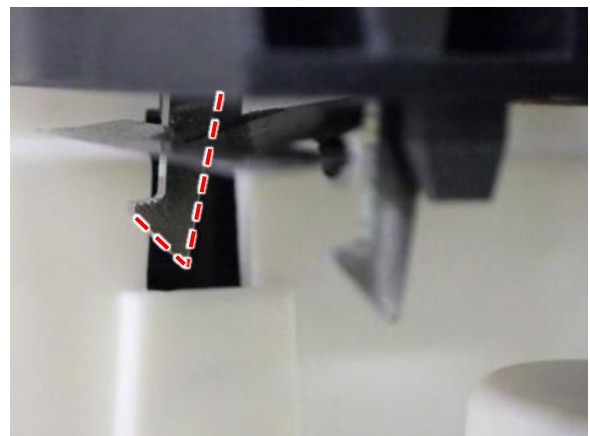
(a): GRIPPER ADJUSTING TOOL 2 (P/N:792-6457)

(b): GRIPPER ADJUSTING TOOL 1 (P/N: 792-6455)

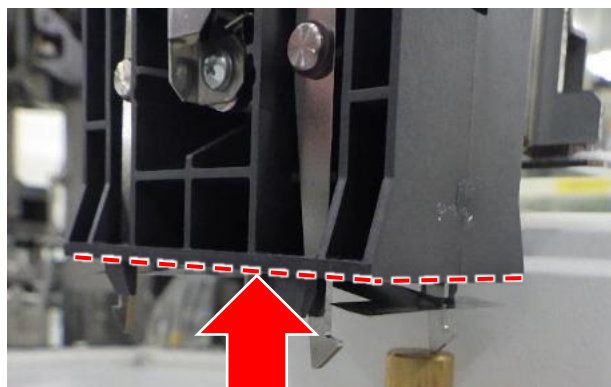
- (2) Select [Set Up].
- (3) Make sure that Gripper claws is the close state.
If not, push up the bottom of Gripper head to switch the claw state.



OK (Close state)



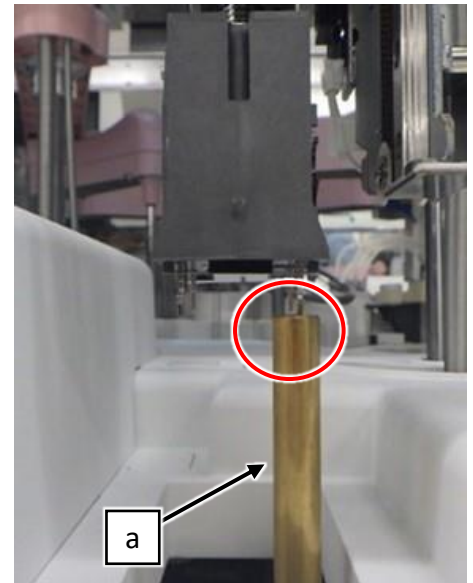
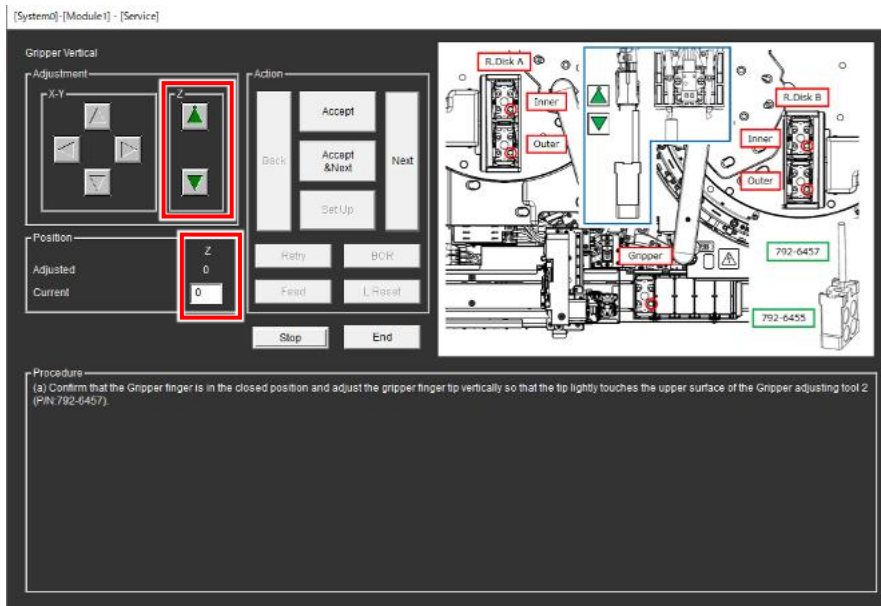
NG (Open state)



- (4) Adjust the height of Gripper head so that Gripper claw touches the upper surface of the pole (a) lightly using the 2 arrow buttons.

(Range: Z: ± 60 pulse)

Note that Gripper claws are in the closed state.

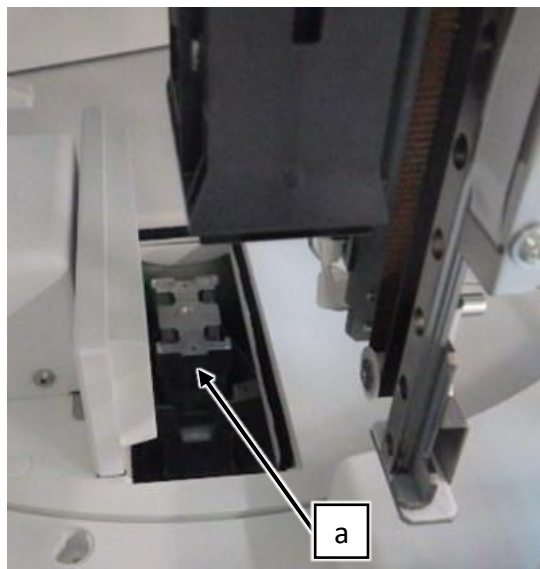


(a): GRIPPER ADJUSTING TOOL 2 (P/N:792-6457)

- (5) Select [Accept & Next].

7.3 R. Disk A Inner Horizontal

- (1) Remove the pole and the tool (a) from the reagent loader.
- (2) Set the tool (a) to Pos.1 of the Reagent Disk A.



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

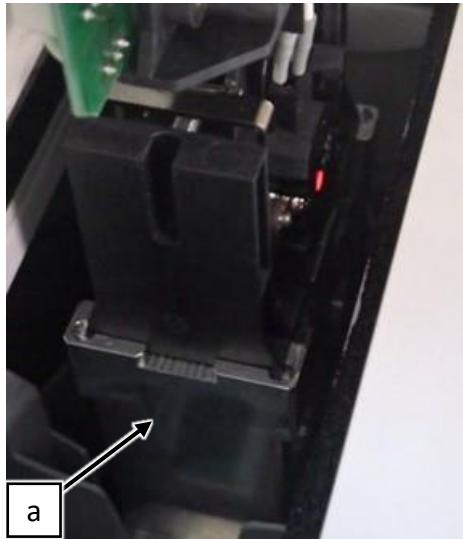
- (3) Select [Set Up].

- (4) Drop the gripper under its own weight and make sure that Gripper head fits into the groove of the tool (a). If it does not drop, gently push the top of gripper and wait for the gripper to fall.

Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.

If it fits smoothly into the groove, select [Next] and proceed to the section 7.4.

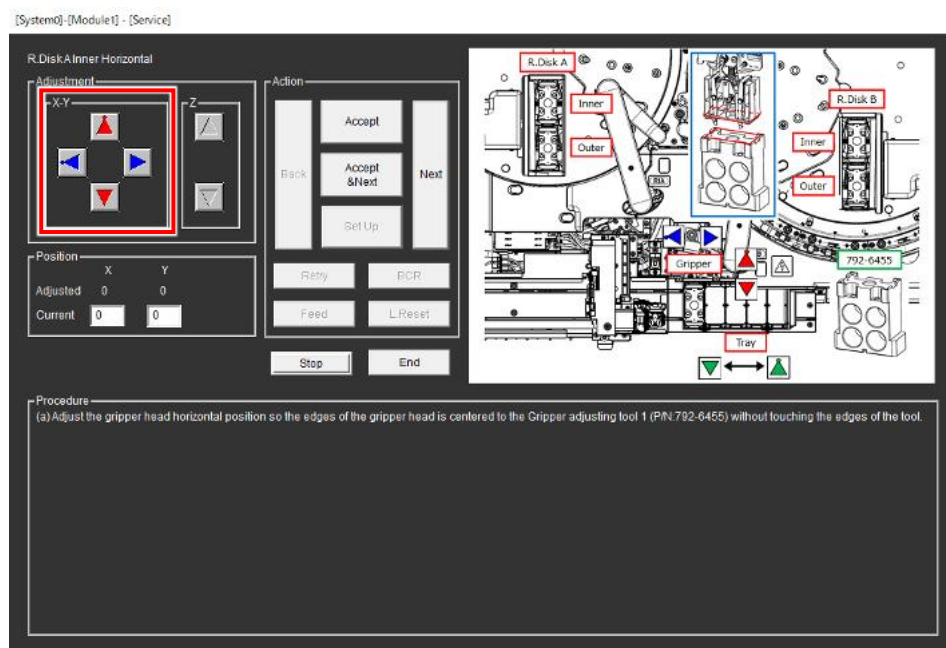
If not, proceed to the step (5).



(a): GRIPPER ADJUSTING TOOL 1 (P/N: 792-6455)

- (5) Adjust Gripper and the reagent loader position so that Gripper head fits smoothly into the tool groove using 4 arrow buttons.

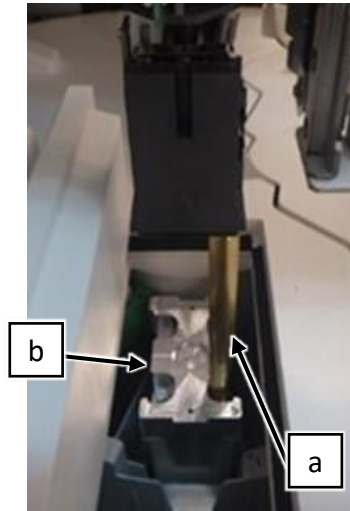
Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.



- (6) Select [Accept & Next].

7.4 R. Disk A Inner Vertical

- (1) Set the pole (a) on the tool (b) hole in front of the right.



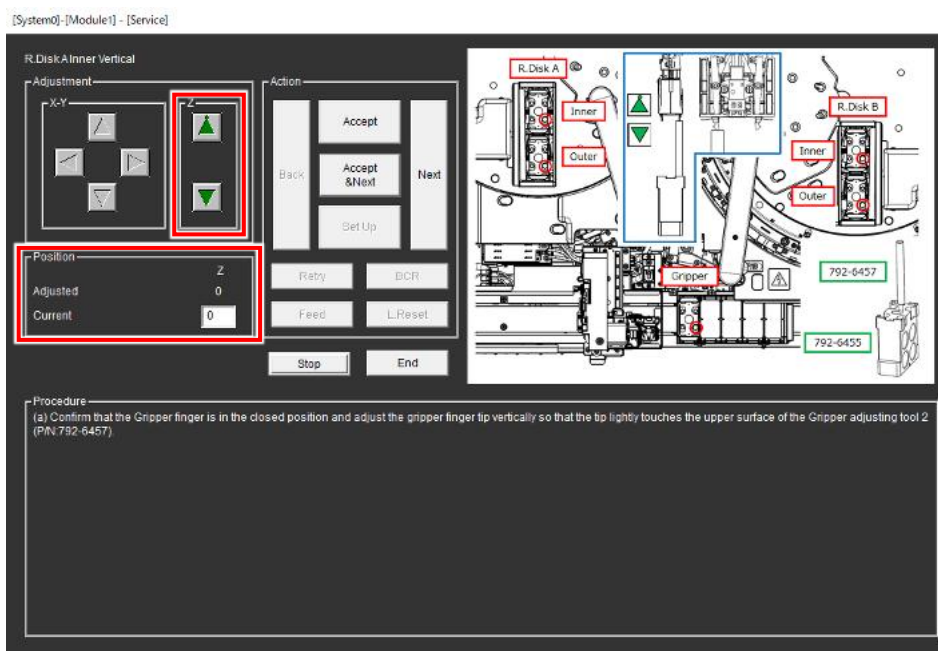
(a): GRIPPER ADJUSTING TOOL 2 (P/N:792-6457)

(b): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (2) Select [Set Up].
- (3) Make sure that Gripper claws is the close state.
If not, push up the bottom of Gripper to switch Gripper state.
- (4) Adjust the height of Gripper head so that Gripper claw touches the upper surface of the pole (a) lightly using the 2 arrow buttons.

(Range: Z: ± 60 pulse)

Note: Gripper claws are in the closed state.

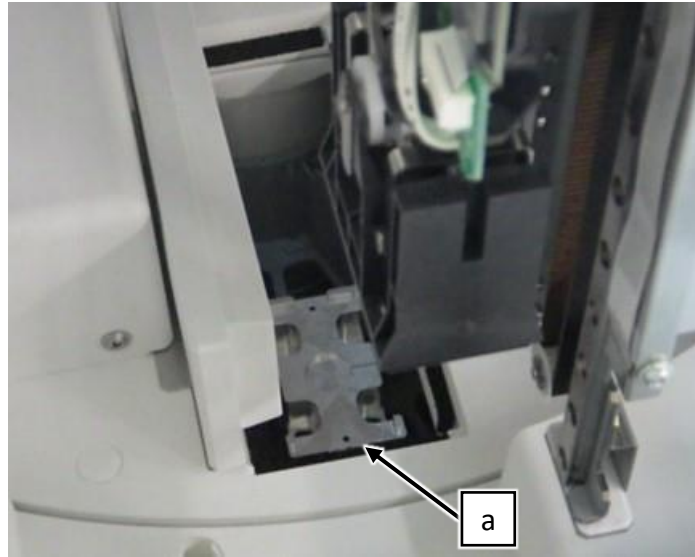


(a): GRIPPER ADJUSTING TOOL 2 (P/N:792-6457)

- (5) Select [Accept & Next].

7.5 R. Disk A Outer Horizontal

- (1) Remove the pole and the tool (a) from Pos.1 of the Reagent Disk A.
(2) Set the tool (a) to Pos.11 of the Reagent Disk A.



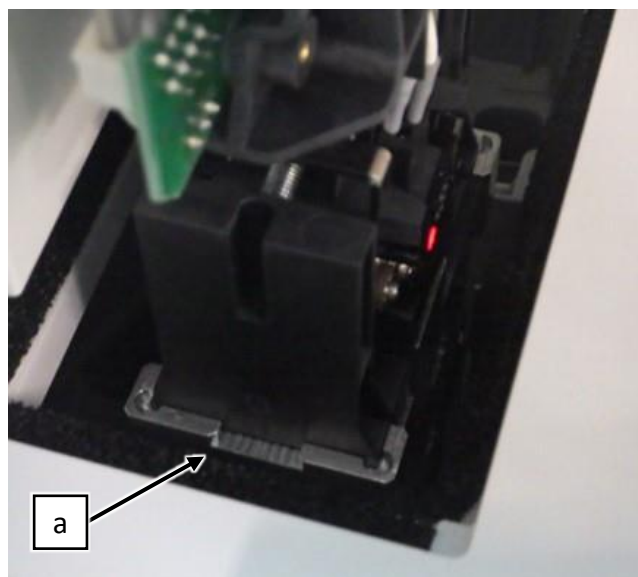
(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (3) Select [Set Up].
(4) Drop the gripper under its own weight and make sure that Gripper head fits into the groove of the tool (a). If it does not drop, gently push the top of gripper and wait for the gripper to fall.

Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.

If it fits smoothly into the groove, select [Next] and proceed to the section 7.6.

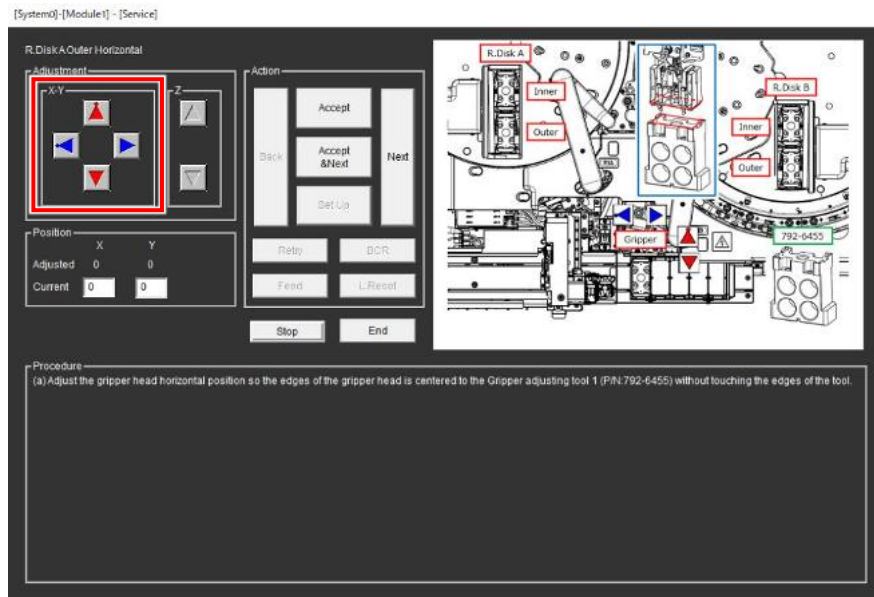
If not, proceed to the step (5).



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (5) Adjust Gripper position so that Gripper head fits smoothly into the tool groove using the 4 arrow buttons.

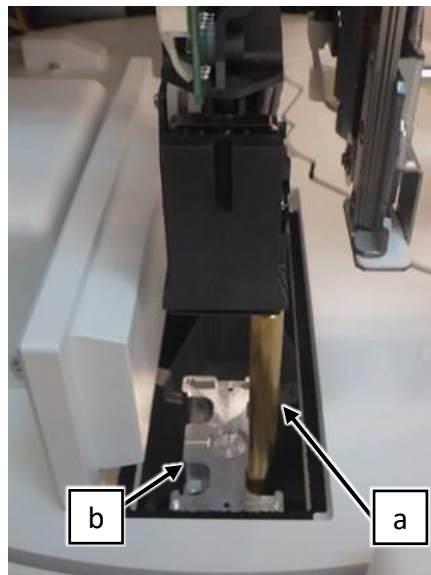
Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.



- (6) Select [Accept & Next].

7.6 R. Disk A Outer Vertical

- (1) Set the pole (a) on the tool (b) hole in front of the right.



(a): GRIPPER ADJUSTING TOOL 2 (P/N:792-6457)

(b): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

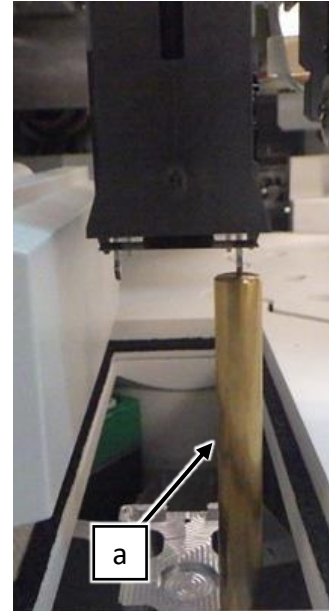
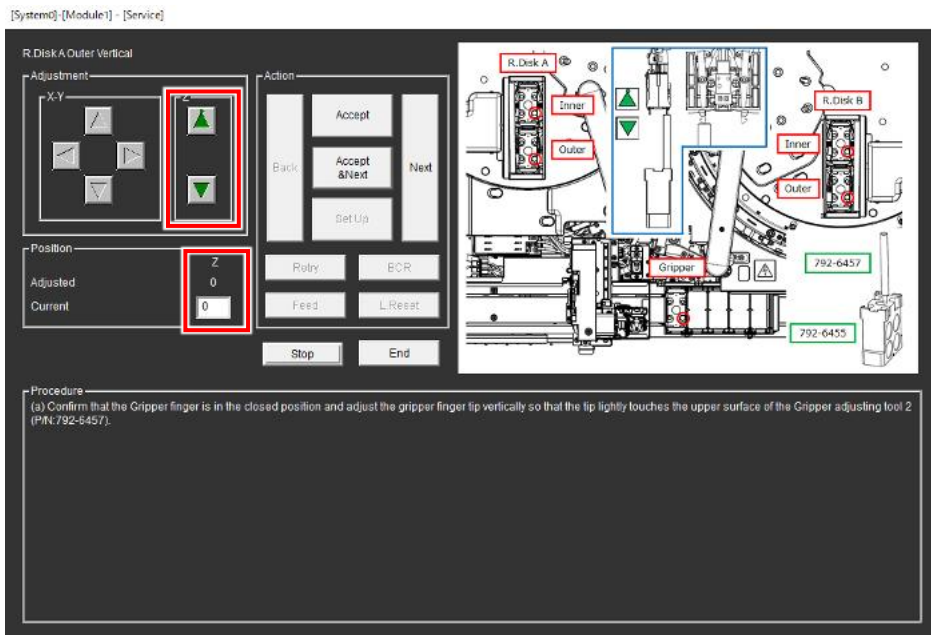
- (2) Select [Set Up].
(3) Make sure that Gripper claws is the close state.

If not, push up the bottom of Gripper to switch the claw state.

- (4) Adjust the height of Gripper head so that Gripper claw touches the upper surface of the pole (a) lightly using the 2 arrow buttons.

(Range: Z: ± 60 pulse)

Note that Gripper claws are in the closed state.

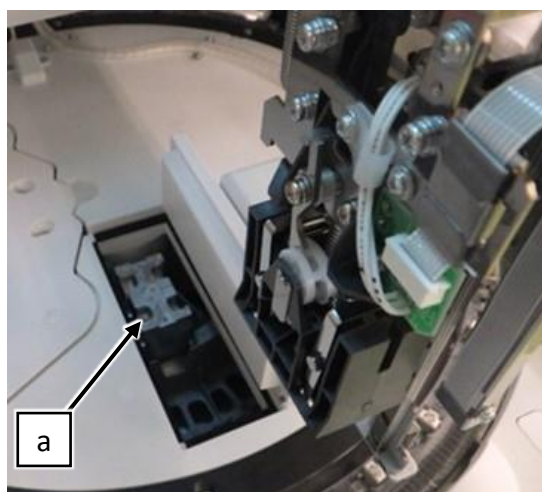


(a): GRIPPER ADJUSTING TOOL 2 (P/N:792-6457)

- (5) Select [Accept & Next].

7.7 R. Disk B Inner Horizontal

- (1) Remove the pole and the tool (a) from the reagent loader.
- (2) Set the tool (a) to Pos.1 of the Reagent Disk B.



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

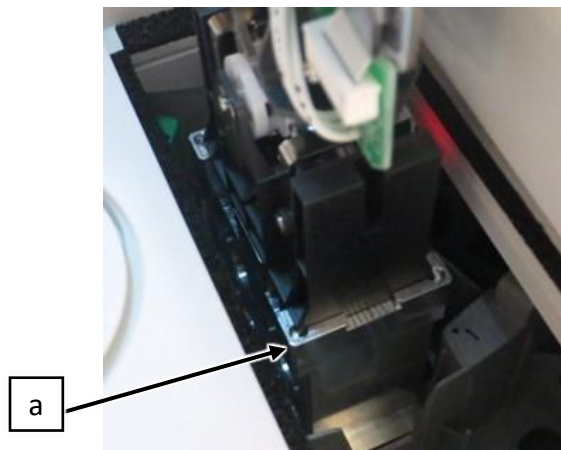
- (3) Select [Set Up].
- (4) Drop the gripper under its own weight and make sure that Gripper head fits into the groove of the tool (a). If it does

not drop, gently push the top of gripper and wait for the gripper to fall.

Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.

If it fits smoothly into the groove, select [Next] and proceed to the section 7.8.

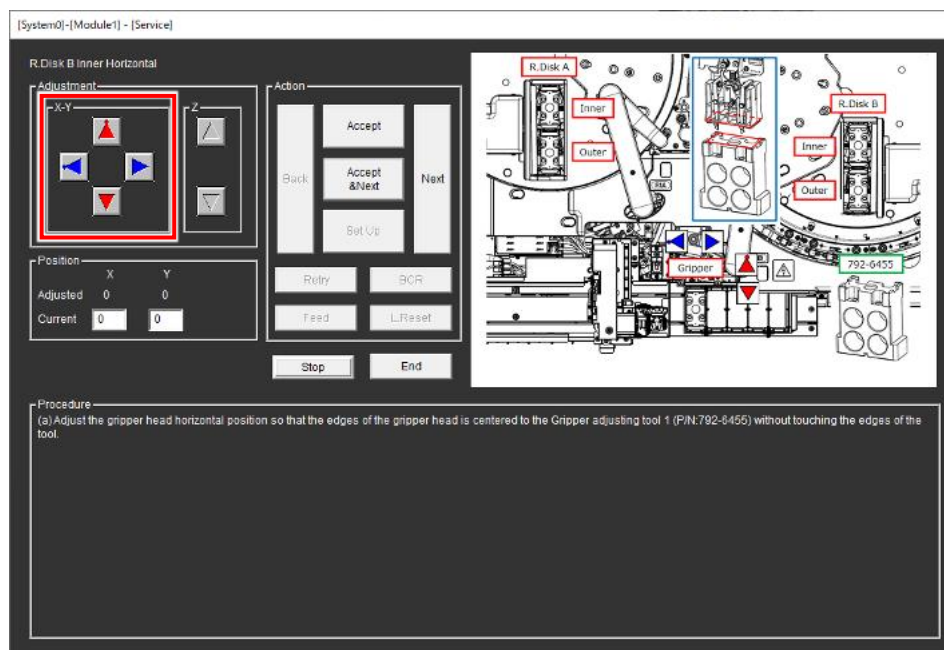
If not, proceed to the step (5).



(a): GRIPPER ADJUSTING TOOL 1 (P/N: 792-6455)

- (5) Adjust Gripper and Reagent Try position so that Gripper head fits smoothly into the tool groove using the 4 arrow buttons.

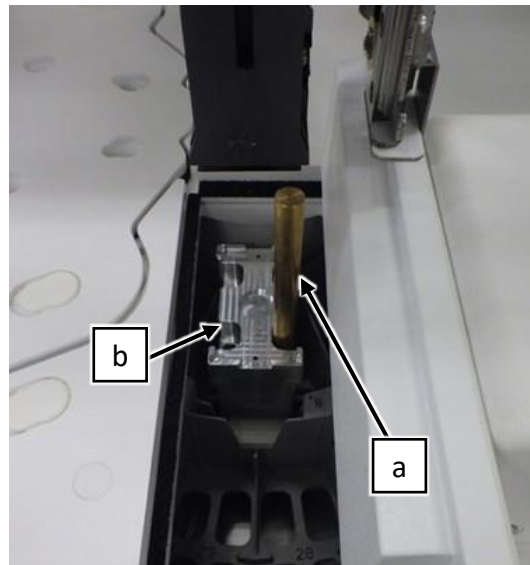
Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.



- (6) Select [Accept & Next].

7.8 R. Disk B Inner Vertical

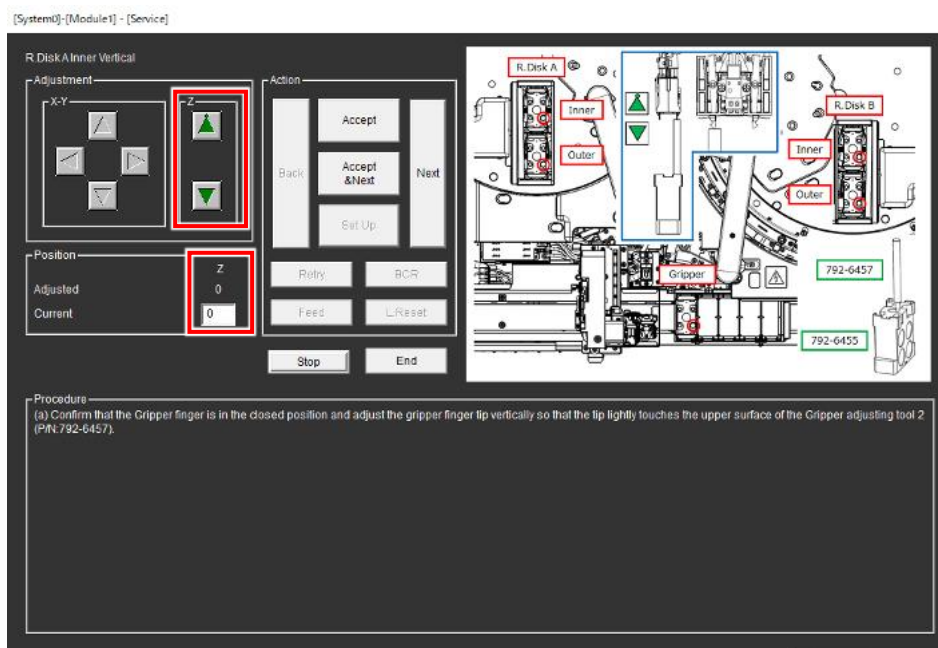
- (1) Set (a) on the tool (b) hole in front of the right.



(a): GRIPPER ADJUSTING TOOL 2 (P/N:792-6457)

(b): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (2) Select [Set Up].
- (3) Make sure that Gripper claws is the close state.
If not, push up the bottom of Gripper to switch Gripper state.
- (4) Adjust the height of Gripper head so that Gripper claw touches the upper surface of the pole (a) lightly using the 2 arrow buttons. (Range: Z: ± 60 pulse)
Note: Gripper claws are in the closed state.

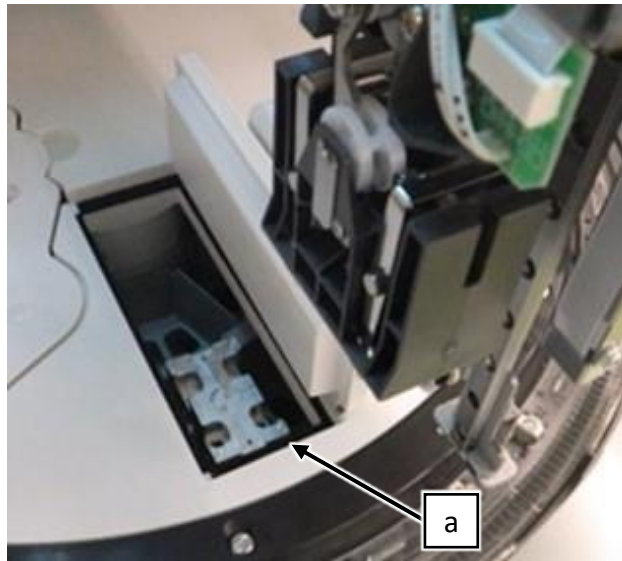


(a): GRIPPER ADJUSTING TOOL 2 (P/N:792-6457)

- (5) Select [Accept & Next].

7.9 R. Disk B Outer Horizontal

- (1) Remove the pole and the tool (a) from Pos.1 of the Reagent Disk B.
- (2) Set the tool (a) to Pos.11 of the Reagent Disk B.



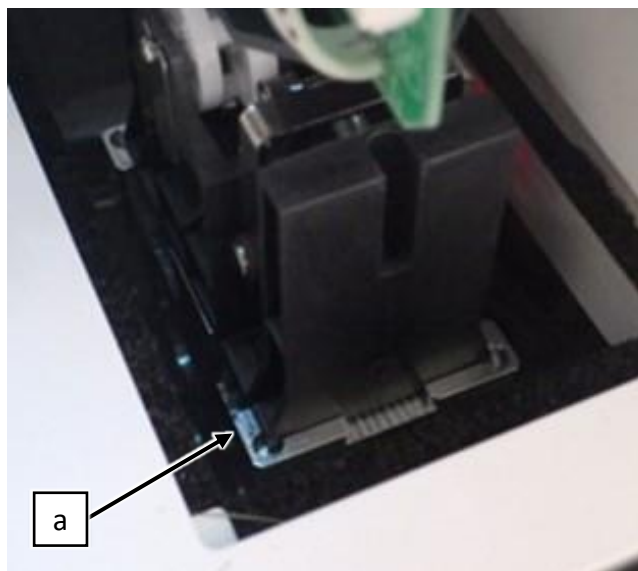
(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (3) Select [Set Up].
- (4) Drop the gripper under its own weight and make sure that Gripper head fits into the groove of the tool (a). If it does not drop, gently push the top of gripper and wait for the gripper to fall.

Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.

If it fits smoothly into the groove, select [Next] and proceed to the section 7.10.

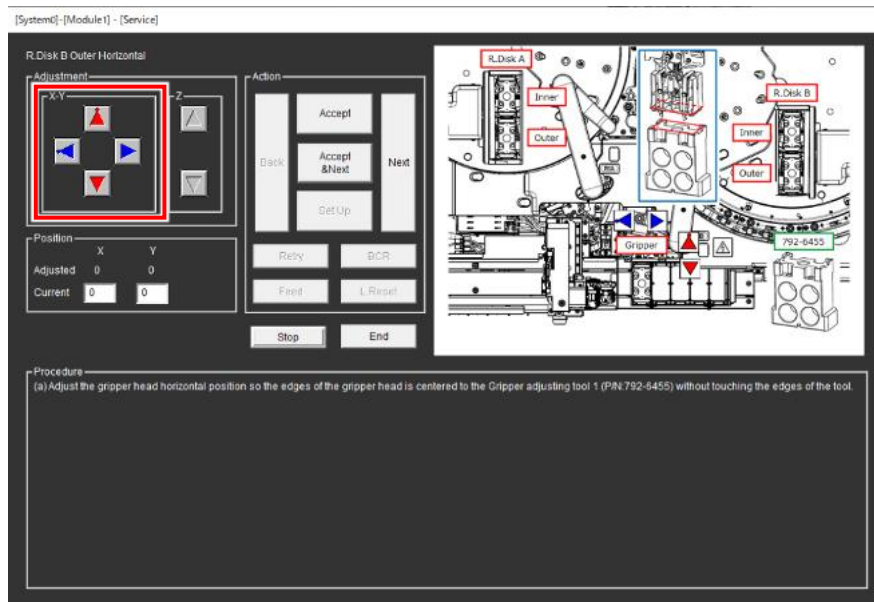
If not, proceed to the step (5).



(a): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

- (5) Adjust Gripper position so that Gripper head fits smoothly into the tool groove using the 4 arrow buttons.

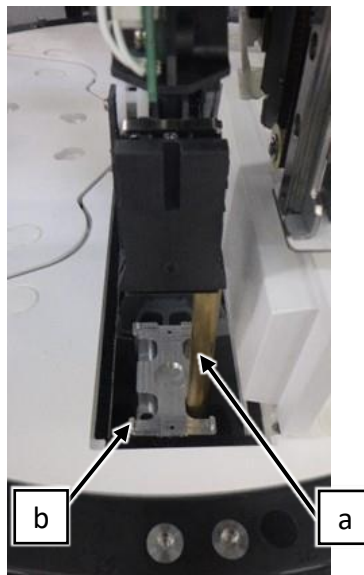
Note: Do not pull Gripper head itself when lowering, because Gripper head becomes an abnormal adjustment condition.



- (6) Select [Accept & Next].

7.10 R. Disk B Outer Vertical

- (1) Set the pole (a) on the tool (b) hole in front of the right.



(a): GRIPPER ADJUSTING TOOL 2 (P/N:792-6457)

(b): GRIPPER ADJUSTING TOOL 1 (P/N:792-6455)

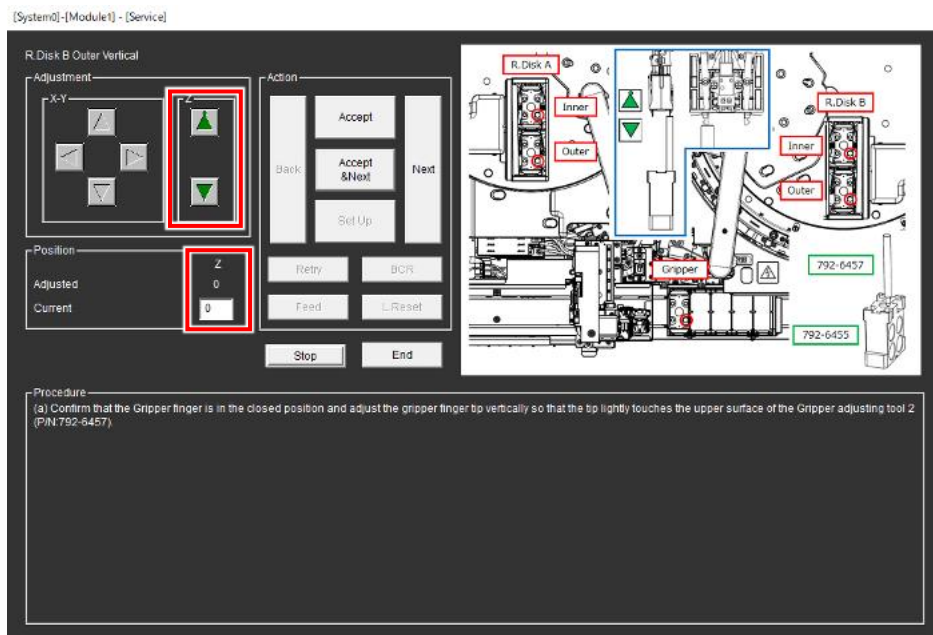
- (2) Select [Set Up].
(3) Make sure that Gripper claws is the close state.

If not, push up the bottom of Gripper to switch the claw state.

- (4) Adjust the height of Gripper head so that Gripper claw touches the upper surface of the pole (a) lightly using the 2 arrow buttons.

(Range: Z: ± 60 pulse)

Note: Gripper claws are in the closed state.



(a): GRIPPER ADJUSTING TOOL 2 (P/N:792-6457)

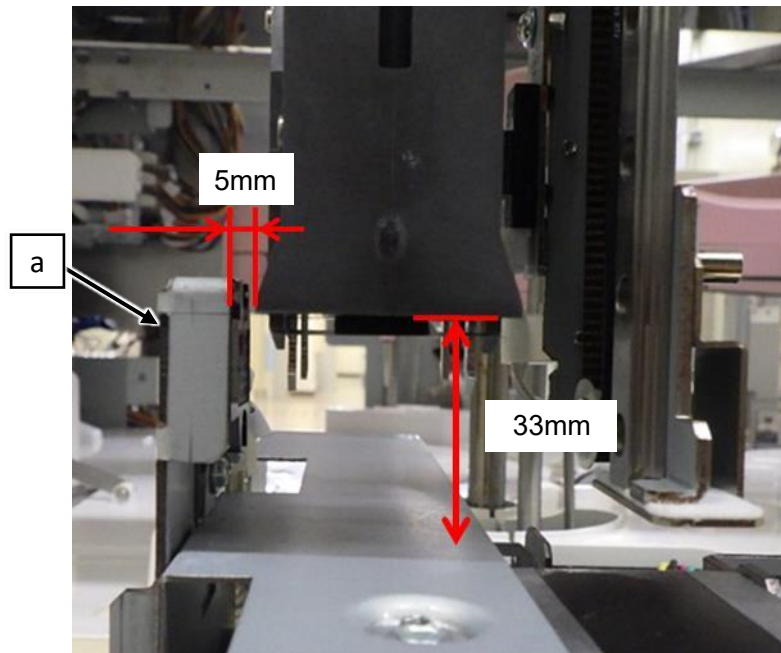
- (5) Select [Accept & Next].

7.11 Gripper Finger Detection

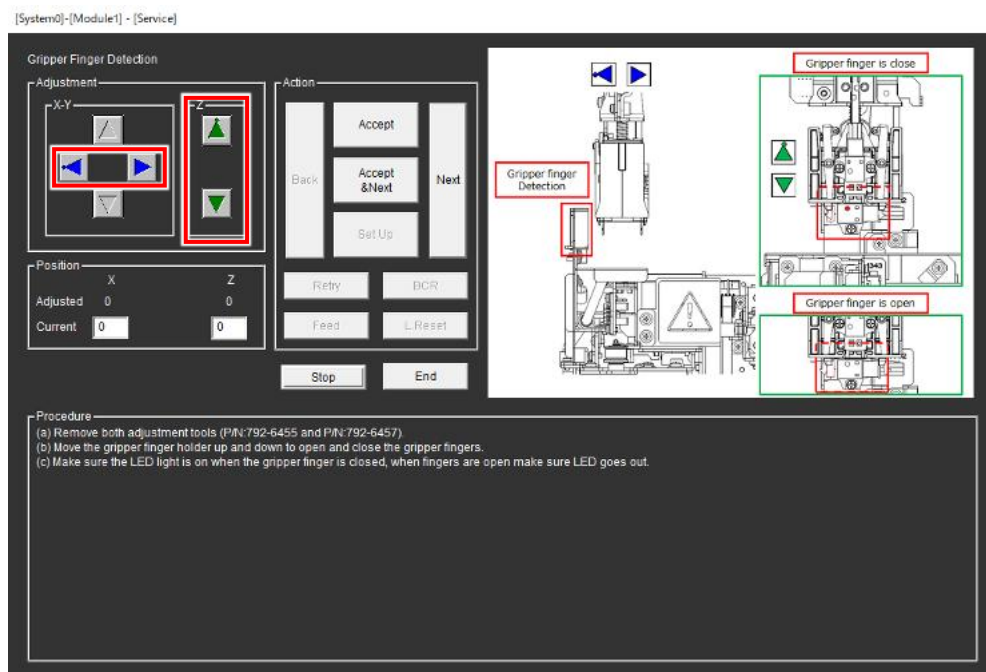
- (1) Remove the pole and the tool from the Reagent Disk B.

S03-08 Reagent Manager Position and Gripper Assy Adjustment

- (2) Make sure that the gap between the PIERCE YZ ASSY (a) and Gripper head is X direction:5mm and Z direction:33mm.
If it is X direction:5mm, Z direction:33mm, proceed to the step (5).
If not, proceed to the step (3).



- (3) Adjust Gripper head position so that the gap between the PIERCE YZ ASSY (a) and Gripper head is X direction:5mm, Z direction:33mm using the 2 arrow buttons.

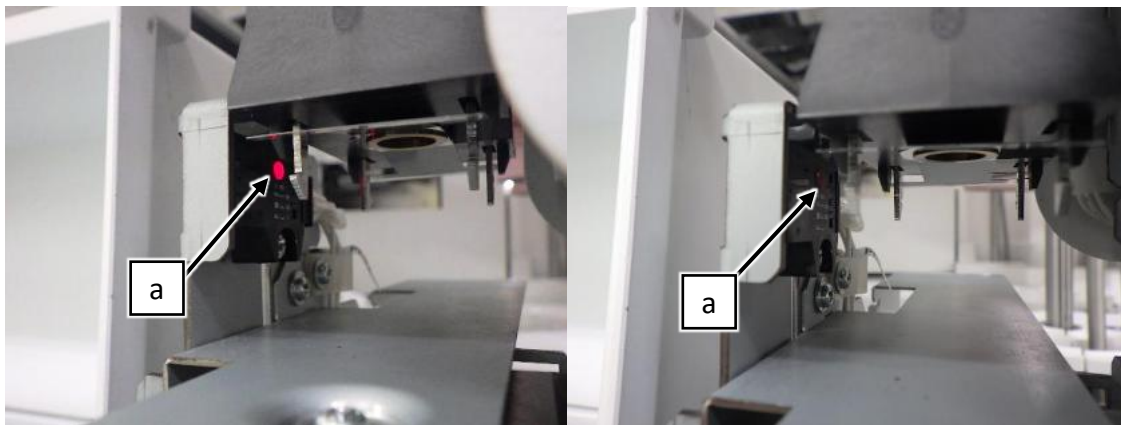
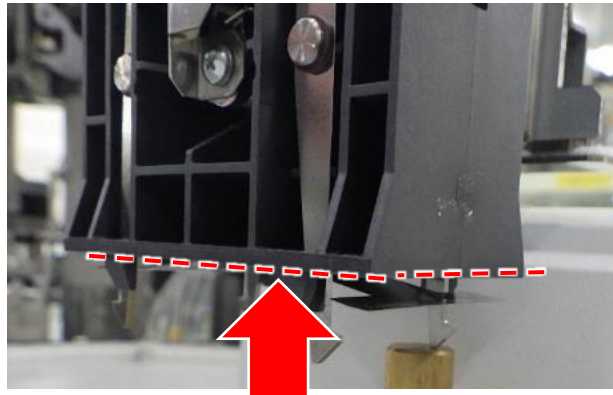


- (4) Select [Accept].

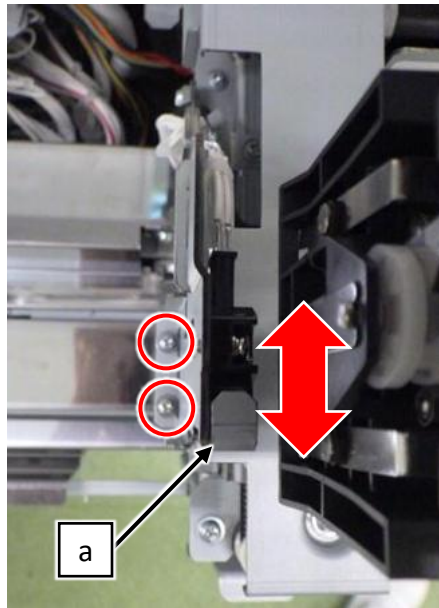
- (5) Push up the bottom to switch the state of Gripper claws and make sure that the LED (a) lights up only when Gripper claws is the close state.

If it does, select [Next] and proceed to the section 7.12.

If not, proceed to the step (6).



- (6) Loosen the 2 screws and adjust the sensor (a) position so that the LED lights up only when Gripper claws is the close state.



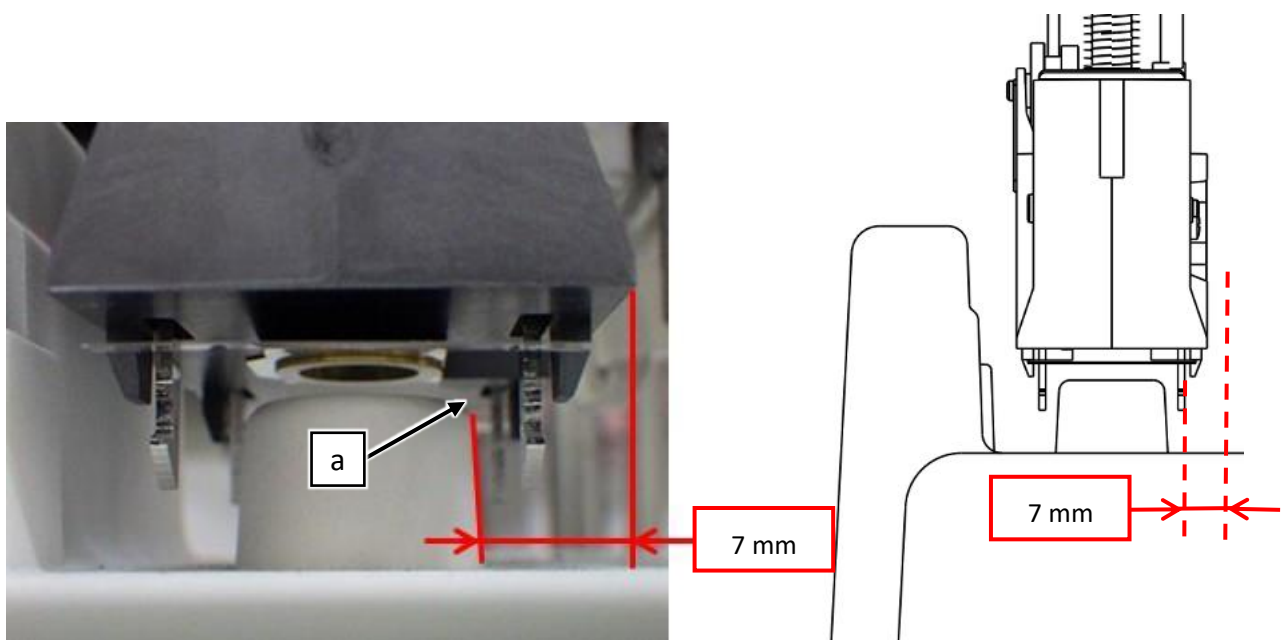
- (7) Tighten the 2 screws.

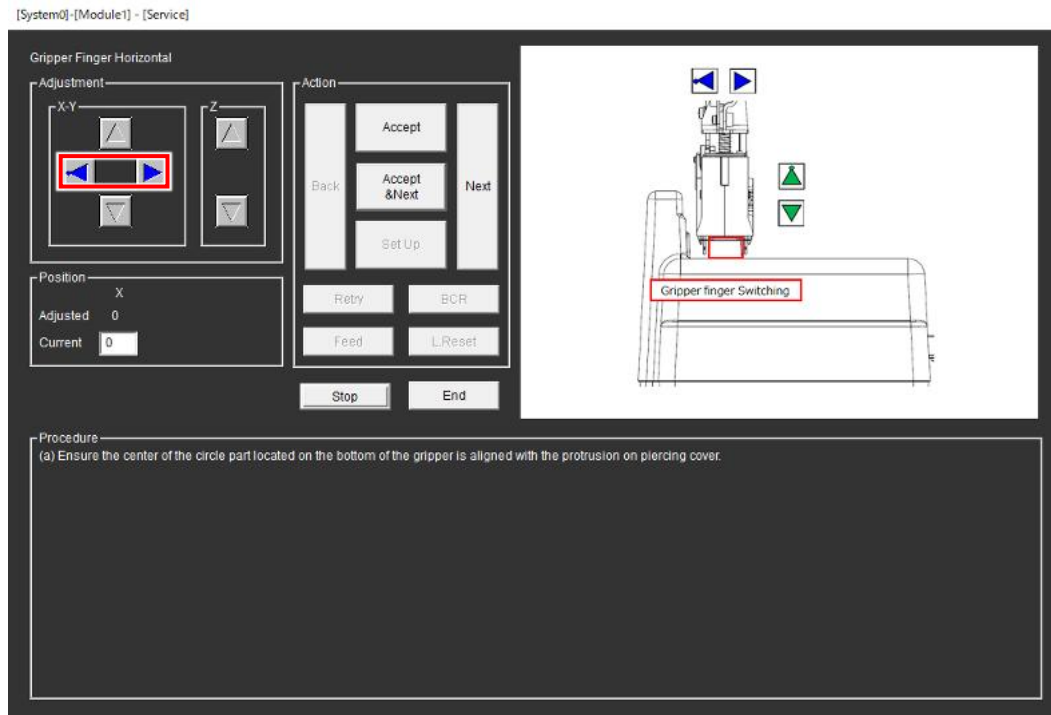
7.12 Gripper Finger Horizontal.

- (1) Attach COVER PIERCE UNIT. (Refer to step 2.1(4).)
- (2) Select [Set Up].
- (3) Rotate the pulley to lower Gripper head and make sure that the distance between COVER PIERCE UNIT (a) projection and Gripper side surface is 7mm.

If it is 7mm, select [Next] and proceed to the section 7.13.

If not, proceed to the step (4).



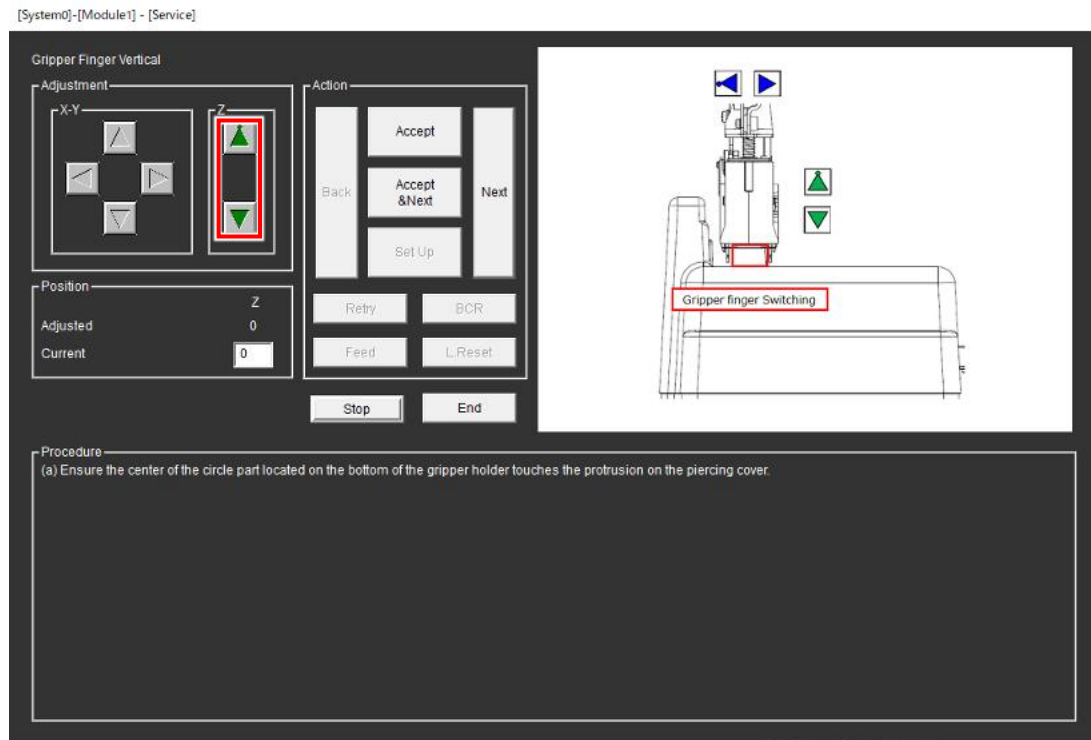
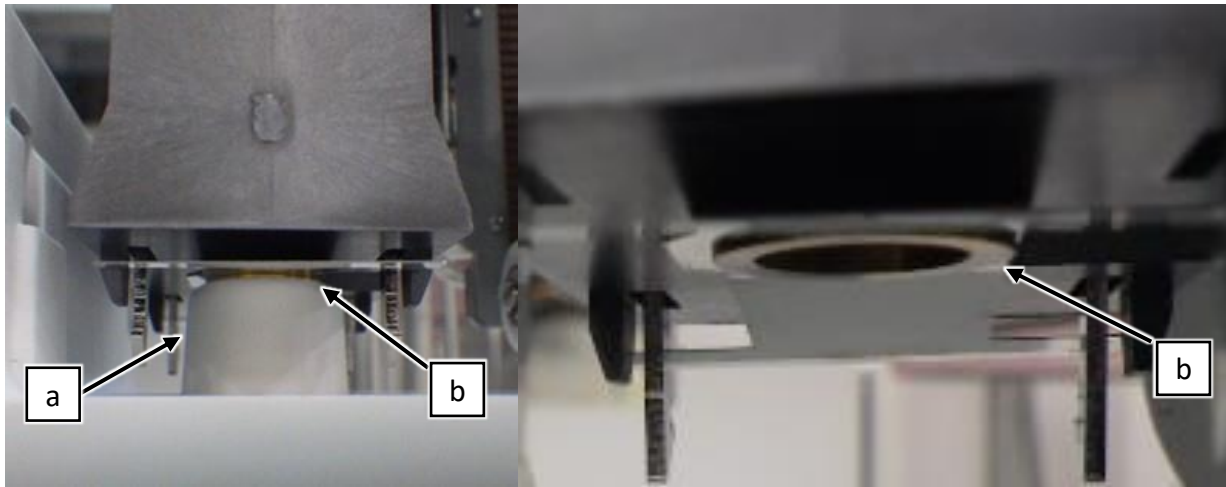


(4) Adjust Gripper position so that using the 2 arrow buttons.

(5) Select [Accept & Next].

7.13 Gripper Finger Vertical

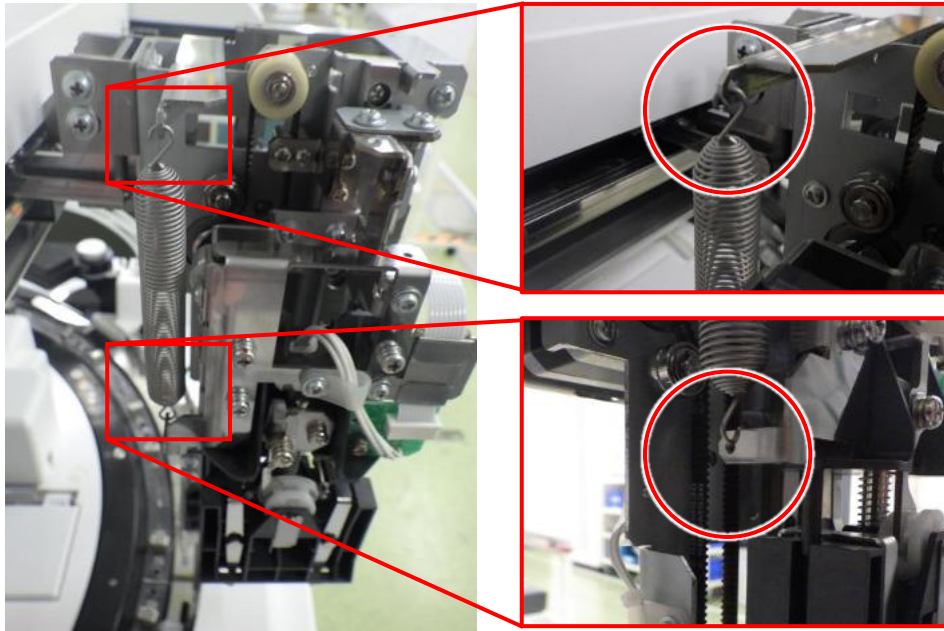
- (1) Select [Set Up].
- (2) Adjust the Gripper head height so that the bottom surface of the metal circle (b) touches the top surface of the projection (a) on the piercing cover lightly using the 2 arrow buttons.



- (3) Select [Accept & Next].
- (4) Select [End].

8 Accomplishing the adjustment

- (1) Turn off the maintenance switch and attach COVER MSW ASSY.
- (2) Install the spring of Gripper.



- (3) Tighten the 2 screws and attach COVER GRIPPER Z 2.
- (4) Tighten the 2 screws and attach COVER GRIPPER Z 1.
- (5) Tighten the 2 screws and attach SW PANEL ASSY.
- (6) Tighten the 2 screws and attach FRONT DOOR then close FRONT DOOR.
- (7) Close COVER TOP F ASSY.

End of Document